Spectral bounds and their limits in Pauli contextuality: exact configurations and a proof of Jonah's Conjecture E

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A general Lean 4 bound, configurations that attain it, and cospectral sign carriers with different completion spectra in every vertex case

Prior work and scope. The spectral floor and its saturation in the three classical families are due to Jonah [J2] and the works identified in Section 1.2. The completion question, the strict law, the classification into sixteen hyperplane cases, the block structure of the carrier, the Jordan representation of the completion, the cone lift and Conjecture E are due to Jonah [J3]. This paper combines a formalization and exact configurations attaining the floor with explicit sign carriers that prove Conjecture E in all nine vertex cases. Case C3 is proved in Lean for every number of qubits from four on; the other eight cases are proved on paper, using two structure theorems of [J3], and checked by exact computation. None of this invalidates the spectral floor.

Abstract

Spectral methods give rigorous bounds on Pauli contextuality, but a spectrum need not retain the information required by a related completion problem. For N qubits, $D = 2^N$, and an arbitrary set S of nonidentity Pauli observables, a Lean 4 development proves Jonah's spectral floor: satisfied minus violated commuting triples is at most $(D-2)|S|^{3/2}/(6\sqrt{D-1})$. Saturating configurations outside the classical families have exact contextuality degrees 75 on 135 four-qubit observables, 480 on 496 five-qubit observables, and 729344 on 8128 seven-qubit observables; all three degrees are proved in Lean with kernel-checked counts. We relate saturation to pure states with a single nonzero Pauli modulus, two-character sets and flat phase states, and classify a specified third-level phase-state family through five qubits.

We then turn to Jonah's strict law: does the spectrum of a sign carrier on a hyperplane section determine the spectrum of the completion operator on the remaining observables? Two sign assignments on a 71-point section of the real four-qubit configuration have cospectral 16×16 carriers, but their 64×64 completion operators have cubic traces 14976 and 15360, and their exact binary completion minima differ. A graph construction extends the trace gap to every $N \geq 4$, with cubic traces that differ by $\frac{3}{2}\,8^{N-1}$; Lean proves this case, C3, for every $N \geq 4$ in the actual Pauli model. Variants over the real, complex and quaternionic numbers and a pair swap for cones give such carriers in all nine vertex cases for every $N \geq 4$, proving Jonah's Conjecture E. Outside case C3 the proof uses two structure theorems of Jonah and is checked by exact computation up to six qubits. Independent exact verifiers, falsification controls and Tsotchke software demonstrations accompany the paper. The work is AI-assisted and is neither externally peer reviewed nor a quantum-hardware experiment.

This journal-format document is derived from the complete paper, version 1.3.0. Original section numbers and cross-references are retained; gaps are intentional. The supplement contains: 10. A running example on the Tsotchke stack, 11. Falsifiability and replication, Appendix A. Reproduction, Appendix B. Witness assignments. References are reproduced in both journal documents.

1. Scope and common question

A noncontextual assignment gives each Pauli observable a fixed value ± 1. Compatible triples impose product constraints, and the contextuality degree counts the fewest constraints any assignment must violate [dB]. A signed sum of Pauli matrices converts this discrete problem into a problem about spectral moments. Its third trace records the satisfied-minus-violated count exactly. Optimizing that moment gives a lower bound on contextuality; special states turn the bound into an exact answer.

A different use of spectral information arises when the assignment is fixed only on a hyperplane section. Completing it on the remaining observables gives a quadratic optimization governed by a second operator. Does the spectrum of the fixed section determine the spectrum of this completion operator? The examples below show that it need not, in every vertex case of Jonah's classification and for every number of qubits from four on: the section spectrum can forget which eigenvalues belong to the eigenspaces of a distinguished Pauli. The completion depends on that partition. Thus a sharp global bound and a failure of spectral reconstruction coexist without contradiction.

1.1 Contributions and verification levels

The first part formalizes the spectral floor for arbitrary induced Pauli configurations and studies its equality cases. The Lean theorem `hoggar_bound` is stated for the actual complex Pauli matrices, with their multiplication phases and context signs. Finite corollaries include the known exact degrees 63 for three qubits and 315 for the real four-qubit quadric, and stronger lower bounds than $L/5$ for several larger classical configurations. The bound itself is prior work [J2]; the formalization and the extension of its application to arbitrary supports are the contributions considered here.

Section 5 gives exact configurations outside the classical families. The four-qubit pivot support S_B has 135 points, 1095 contexts and degree 75; the five-qubit support S_5 has 496 points, 10880 contexts and degree 480. The seven-qubit circuit support has 8128 points, 2824192 contexts and degree 729344. These three degree statements, including their finite counts and witnesses, are proved in Lean; the seven-qubit counts are double sums with 2^{28} terms, evaluated by the kernel on packed sets of codes. Their interpretation as Pauli supports of particular states is checked separately by exact programs. The saturation criterion, quantum-metric interpretation, two-character-set result and classification of specified flat phase states are mathematical and executable results with the boundaries stated in Section 9.

Section 8 concerns Jonah's strict law and Conjecture E [J3]. It first supplies a fully specified four-qubit C3 certificate: two assignments on the 71-point section of the real quadric have equal carrier spectra, different completion spectra (Theorem 14) and different exact binary completion minima (Corollary 16). Lean checks the equal carrier spectra and different completion spectra in the actual Pauli and signed-context interpretation, not merely for an isolated pair of matrices; the printed factorizations and Corollary 16 rest on exact algebraic and executable checks, outside the 42-declaration C3 audit. It then gives an explicit family of pairs with equal carrier spectra and different completion spectra for every $N \geq 4$, built from graphs with one edge for each

nonzero difference (Theorem 17); its Lean statement `c3_strict_law_fails_general` is about the same Pauli matrices, real quadric and context signs as the floor. By a lifting lemma of [J3] this also settles case C11 (Corollary 18). Finally, variants of the construction prove Conjecture E in all nine vertex cases (Theorem 19). That proof is on paper outside case C3, uses Theorems A and B3 of [J3], and is checked by exact computation on the complete configurations for three to six qubits.

The two 135-point configurations must be distinguished. The pivot support S_B is not a quadric and has 1095 contexts and degree 75. The C3 ambient configuration is the real quadric $Q^+(7,2)$, with 1575 contexts and known degree 315 [LGG]; its section leaves 1120 outside contexts. The C3 result concerns completion spectra and does not calculate a new contextuality degree for either set.

Section 9 records the formal boundary for both parts. Sections 10 and 11 give the software demonstrations and falsification checks; Appendices A and B give reproduction commands and witness formats. In the journal layout those four sections form the supplement, retaining their numbers. The full paper and journal forms share the same mathematical statements and named references.

1.2 Relation to prior work

- The contextuality degree and its first values: de Boutray, Holweck, Giorgetti, Masson and Saniga [dB]; Muller, Saniga, Giorgetti, de Boutray and Holweck [M1] (three-qubit degree 63, the split Cayley hexagon of violated contexts); Muller, Saniga, Giorgetti, Holweck and Kelleher [M2] (upper bounds for four to seven qubits, lower bound $L/5$); Saniga et al. [S] (hexagons).
- The exact value 315 for the four-qubit hyperbolic quadric, its 240 optimal sets and a Lean proof of that value: Li, Gupta and Ganesh [LGG].
- The exact values $d(Q^-(7,2)) = 315$, $d(W(7,2)) = 1575$, $d(Q^+(9,2)) = 6975$ (computer-assisted), the operator $A = \sum_v f(v)P_v$ and its Hoggar spectrum $\{21, -3^7\}$: Jonah [J1].
- The spectral floor itself, its Kramers (quaternionic) form for elliptic quadrics, the description of quadrics by the antiunitaries P_cK, the operator identity for A^2 on sets closed under products, and the theorem that among the three families the floor is attained exactly for $W(5,2)$, $Q^+(7,2)$, $Q^-(5,2)$ (and trivially $Q^-(3,2)$): Jonah [J2], with the trace formula also in [J3]. Stronger five-qubit lower bounds ($d(W(9,2)) \geq 28830$, $d(Q^-(9,2)) \geq 6510$) are announced, with proofs held privately, in [J4]. Sets that are not closed under products are named in [J2] as not covered.
- Pure states whose Pauli expectation values are zero or of one modulus: the two-qubit state of Liu, Low and Yin [LLY]; the four-qubit state $(-3, 1, \ldots, 1)/\sqrt{24}$ and the five-qubit state $|s_5\rangle$ with 135 and 496 non-zero values, found numerically by Cieśliński [Ci], who calls a (possibly mixed) state *Pauli flat* when all of its Pauli expectation values have one modulus and lists these pure states under the relaxed condition; real four-qubit states of this kind as the optima of $Q^+(7,2)$ [J4]. Kang [K] proves that no four-qubit state, and no state of an odd number $N > 1$ of qubits, has all squared Pauli expectation values in $\{0, 2^{-N}\}$. None of [LLY, Ci, K] discusses contextuality, phase states, or more than five qubits.
- Diagonal gates of the Clifford hierarchy and their phase polynomials: Cui, Gottesman and Krishna [CGK]. $\mathbb{Z}_4$-valued quadratic forms and their character sums: [CCKS]. Two-character sets, two-weight codes and strongly regular graphs: Calderbank and Kantor [CK]. The bent functions of Maiorana and McFarland: [McF].
- The scalar inequality behind the floor is classical: it is Okumura's lemma [O] (equivalently the maximal sample skewness, Wilkins [W]).

- The Hoggar lines and the cubic purity condition for quantum states: Hoggar [H], Stacey [St], Jones and Linden [JL]. Bent functions and their difference sets: Rothaus [R]. Point sets with the hyperplane intersection numbers of a quadric (quasi-quadrics): De Clerck, Hamilton, O'Keefe and Penttila [DHOP].

The spectral-floor derivation in Sections 3, 4 and 6 was developed before the project identified [J1]–[J4]; the overlap is attributed here. The exhaustive computations behind the exact values of [J1] were not rerun. The assignments found in this project are consistent with those values, which remain external computer-assisted claims.

For completion spectra, [J3] gives the strict-law question, its Jordan representation, the case table and Conjecture E. Its inspected v0.5 proves the conjecture for C1 and C6 by a computer-assisted base case and a lifting lemma, reports exact pairs for several other cases at four and five qubits, finds the strict law true for all sign carriers in C5, C8, C12, C14 and C15 at three qubits, reports the three-qubit C3 phenomenon, and leaves C3 for $N \geq 4$ open after random searches at four and five qubits found no pair. Section 8 addresses $N = 4$ and then every vertex case. The literature check read that primary PDF in full, examined related deposits, and on 10 November 2026 ran 54 queries on Zenodo, arXiv, OpenAlex, DataCite, GitHub and the open web for the conjecture, case C3, the graph construction of Section 8.6 and the numbers of its tables. No other treatment of case C3 for $N \geq 4$, or of Conjecture E, was identified in that bounded search. This is a qualitative comparison of evidence, not a statistical meta-analysis or a guarantee of historical priority; unpublished work, in particular by the author of [J3], cannot be excluded. The source manuscript is dated 20 October 2026, its deposit is dated 21 October, and its latest-version endpoint returned v0.5 when inspected on 9 and 10 November.

2. Setting

Pauli operators. Let $D = 2^N$. A *point* is a nonzero vector $v = (a \mid c) \in \mathbb{F}_2^{2N}$. It labels the Hermitian operator

$$P_v \;=\; i^{\,|a \wedge c|}\, X^a Z^c,$$

a tensor product of I, X, Y, Z (the exponent $|a \wedge c|$ counts the factors equal to $Y = iXZ$). For numeric labels use $a + 2^N c$, interpreting each binary vector as an integer with bit zero at the rightmost tensor factor. Words are printed in tensor-factor order from left to right. Thus four-qubit label 128 is $ZIII$. This convention is also used for the C3 witnesses in Section 8. $P_0 = I$, $P_v^2 = I$, and $\operatorname{tr} P_v = 0$ for $v \neq 0$. Products close up to a phase,

$$P_u P_v = \tau(u,v)\, P_{u+v}, \qquad \tau(u,v) = i^{\,k_u + k_v + 3k_{u+v}}\, (-1)^{c_u \cdot a_v}, \quad k_v = |a_v \wedge c_v|,$$

and with the symplectic form $\langle u, v\rangle = c_u \cdot a_v + c_v \cdot a_u$ one has $\tau(v,u) = (-1)^{\langle u,v\rangle}\tau(u,v)$ and $\tau(u,v)^2 = (-1)^{\langle u,v\rangle}$. So P_u and P_v commute exactly when $\langle u, v\rangle = 0$, and then $\tau(u,v) = s(u,v) \in \{+1, -1\}$.

Contexts. A *context* (a line of the symplectic polar space $W(2N-1, 2)$) is a set $\ell = \{u, v, u+v\}$ of three distinct points with $\langle u, v\rangle = 0$. Its three operators commute and $P_u P_v P_{u+v} = s_\ell\, I$ with $s_\ell = s(u,v)$, which does not depend on the order.

Configurations and assignments. Fix a set S of points and let $\mathcal{L}(S)$ be the set of all contexts contained in S, $L_S = |\mathcal{L}(S)|$. S is *closed* if, for distinct labels $u, v \in S$ with $\langle u, v\rangle = 0$, the label $u + v$ also belongs to S: closure of commuting Pauli products up to sign, excluding the identity product of a label with itself. An assignment $f : S \to \{+1, -1\}$ *violates* ℓ when $f(u)f(v)f(u+v) \neq s_\ell$. Put

$$\beta_S(f) = \sum_{\ell \in \mathcal{L}(S)} s_\ell\, f(u)f(v)f(u+v) = L_S - 2\,\mathrm{viol}(f), \qquad d(S) = \min_f \mathrm{viol}(f).$$

$d(S)$ is the contextuality degree [dB]. For the noncontextuality inequality $\chi = \sum_\ell s_\ell \langle P_u P_v P_{u+v}\rangle \le L_S - 2d(S)$ quantum mechanics gives $\chi = L_S$ in every state [C, LGG].

Three families. The *full space* is $S = \mathbb{F}_2^{2N}\setminus\{0\}$, with D^2-1 points and $L = (D^2-1)(D^2-4)/12$ contexts. Let $q_0(v) = |a \wedge c| \bmod 2$ for $v = (a \mid c)$. For $w \in \mathbb{F}_2^{2N}$ set $Q_w = \{v \neq 0 : q_0(v) + \langle w, v\rangle = 0\}$. Q_w is a *hyperbolic* quadric $Q^+(2N-1,2)$ if $q_0(w) = 0$ and an *elliptic* quadric $Q^-(2N-1,2)$ if $q_0(w) = 1$. With K complex conjugation in the computational basis, $\theta_w = P_w K$ is antiunitary, $\theta_w P_v \theta_w^{-1} = (-1)^{q_0(v)+\langle w,v\rangle} P_v$ and $\theta_w^2 = (-1)^{q_0(w)}$. So the quadrics are the sets of Pauli observables fixed by a Pauli antiunitary: a real structure ($\theta^2 = 1$) for hyperbolic quadrics and a quaternionic structure ($\theta^2 = -1$) for elliptic ones [J2]. Q_0 is the set of real symmetric Pauli matrices (an even number of Y factors). All three families are closed.

3. The spectral floor

Theorem 1 (spectral floor; Jonah [J2] for the three families; formalized here for arbitrary S). Let $N \ge 1$, let S be any set of points and $f : S \to \{\pm 1\}$. Then

$$\beta_S(f) \;\le\; \frac{(D-2)\,|S|^{3/2}}{6\sqrt{D-1}}, \qquad \text{hence} \qquad d(S) \;\ge\; \frac{L_S}{2} - \frac{(D-2)\,|S|^{3/2}}{12\sqrt{D-1}}.$$

Proof. Let $A = \sum_{v\in S} f(v) P_v$, a Hermitian $D \times D$ matrix.

(i) Three traces. $\mathrm{tr}\, A = 0$ because $0 \notin S$. Since $\mathrm{tr}(P_u P_v) = D\,\delta_{uv}$ and $f(v)^2 = 1$, $\mathrm{tr}\, A^2 = D\,|S|$. Since $\mathrm{tr}(P_u P_v P_w) = D\,\tau(u,v)\,[w = u+v]$,

$$\mathrm{tr}\, A^3 = D \sum_{\substack{u,v\in S\\ u+v\in S}} f(u)f(v)f(u+v)\,\tau(u,v).$$

The pairs with $\langle u, v\rangle = 1$ cancel against their transposes, because $\tau(v,u) = -\tau(u,v)$. The pairs with $\langle u, v\rangle = 0$ are the ordered pairs of distinct points of a context ($u = v$ would give $u + v = 0 \notin S$), six for each context, each contributing $s_\ell f(u)f(v)f(u+v)$. Therefore $\mathrm{tr}\, A^3 = 6D\,\beta_S(f)$.

(ii) A scalar inequality (Okumura [O]). Let $x_1, \dots, x_n$ be real with $\sum x_i = 0$ and $\sum x_i^2 = nm$, $n \ge 2$, and let $a = \sqrt{(n-1)m}$. By the Cauchy–Schwarz inequality applied to the other $n-1$ numbers, $x_i^2 \le (n-1)(nm - x_i^2)$, so $x_i \le a$ for every i. Hence $(x_i - a)\big((n-1)x_i + a\big)^2 \le 0$. Summing over i and using the two constraints,

$$(n-1)\sum_i x_i^3 \;\le\; n\,m\,(n-2)\,a.$$

(iii) Conclusion. Apply (ii) to the eigenvalues of A with $n = D$, $m = |S|$: $(D-1)\cdot 6D\beta \le D\,|S|\,(D-2)\sqrt{(D-1)|S|}$. ∎

Three remarks. First, no closure property of S is used: products that leave S do not contribute to the trace. Second, the hypothesis $f = \pm 1$ enters only through $\mathrm{tr}\, A^2 = D|S|$, so the argument bounds the cubic form $T(x) = \sum_\ell s_\ell x_u x_v x_w$ on the sphere $\sum x_v^2 = |S|$. Third, the bound applies to the induced configuration only, that is, with *all* contexts contained in S.

Theorem 2 (quaternionic form; Jonah [J2]). Let $N \ge 2$. If every P_v, $v \in S$, commutes with an antiunitary θ with $\theta^2 = -1$ (for instance, S lies in an elliptic quadric), then

$$\beta_S(f) \;\le\; \frac{(D-4)\,|S|^{3/2}}{6\sqrt{2(D-2)}}.$$

Proof. A is a real combination of operators commuting with θ, so $\theta A = A\theta$. If $Ax = \lambda x$ then $A\theta x = \lambda\theta x$, and $\langle x, \theta x\rangle = 0$ because $\langle \theta x, \theta y\rangle = \overline{\langle x, y\rangle}$ gives $\langle x, \theta x\rangle = \overline{\langle \theta x, \theta^2 x\rangle} = -\langle x, \theta x\rangle$. Every eigenspace is therefore an orthogonal sum of planes $\{x, \theta x\}$ and every eigenvalue has even multiplicity (Kramers). Apply (ii) to one value from each of the $D/2$ pairs, counted with pair multiplicity, with the same m. ■

Theorem 2 is not part of the Lean development; it is checked numerically (Section 11).

Closed forms. With the standard counts, Theorems 1 and 2 give for the three families the same expression in different arguments. Writing $\varphi(D) = \frac{1}{2} - \sqrt{D+1}/(D+2)$,

the fraction column below is understood only when $L > 0$; configurations without contexts have degree zero, and no violated fraction is defined for them.

family	points	contexts L	violated fraction at least
full space $W(2N-1,2)$	D^2-1	$(D^2-1)(D^2-4)/12$	$\varphi(D)$
elliptic quadric $Q^-(2N-1,2)$	$(D+1)(D-2)/2$	$(D+1)(D-2)(D+2)(D-4)/48$	$\varphi(D)$
hyperbolic quadric $Q^+(2N-1,2)$	$(D-1)(D+2)/2$	$(D-1)(D+2)(D-2)(D+4)/48$	$\varphi(D/2)$

The three cases are the complex, quaternionic and real members of Dyson's threefold way; the real family at N qubits has the same floor as the complex family at $N-1$ qubits.

4. Saturation in the classical families

Theorem 3 (equality; Jonah [J2, Theorem 6] for the three families). Suppose $S \neq \varnothing$. Equality holds in Theorem 1 if and only if there is a unit vector ψ with

$$\langle\psi|P_v|\psi\rangle = f(v)\sqrt{\frac{D-1}{|S|}} \quad (v \in S), \qquad \langle\psi|P_v|\psi\rangle = 0 \quad (v \notin S,\ v \neq 0).$$

Proof. Equality in (ii) forces $x_i \in \{a, -a/(n-1)\}$ for every i, and the zero sum forces exactly one value a. So $A = (a-b)\,|\psi\rangle\langle\psi| + b\,I$ with $b = -a/(D-1)$; reading off Pauli coefficients gives the claim, and the converse is the same computation backwards. ■

We call such a ψ *flat on* S, after [Ci]. For the full space the condition says that ψ is a fiducial vector of a SIC covariant under the N-qubit Pauli group. Such vectors exist only for $N = 1$ and $N = 3$ (Godsil and Roy [GR]; an elementary proof through the integrality of the floor is in [J2]): for $N = 3$ they are the Hoggar lines.

The three saturated cases. The Hoggar fiducial $\psi_H \propto (-1+2i, 1, 1, 1, 1, 1, 1, 1)$ has $\langle P_v\rangle = \pm 1/3$ for all 63 three-qubit Pauli observables. We verified the following by exact counting in four independent implementations (Section 11); the values themselves are known [M1, dB, LGG, J2].

configuration	assignment	violated	floor	spectrum of A
$W(5,2)$: 63 points, 315 contexts	signs of $\langle\psi_H\|P_v\|\psi_H\rangle$	63	63	$\{21, (-3)^7\}$

configuration	assignment	violated	floor	spectrum of A
$Q^-(5,2)$: 27 points, 45 contexts	the same signs, restricted	9	9	$\{9^2,(-3)^6\}$
$Q^+(7,2)$: 135 points, 1575 contexts	signs of the realified $\psi_H \in \mathbb{R}^{16}$	315	315	$\{45,(-3)^{15}\}$

In the third row, $\mathbb{C}^8$ is read as $\mathbb{R}^{16}$; the resulting real four-qubit state has expectation $\pm 1/3$ on all 135 real Pauli observables and 0 on the 120 imaginary ones (this description of the optimal assignments of $Q^+(7,2)$ as sign patterns of real flat states is announced in [J4]; the state $(-3,1,\dots,1)/\sqrt{24}$ of [Ci] is of this kind). The violated contexts form, respectively, a classically embedded split Cayley hexagon (63 lines, three per point, incidence graph of diameter 6 and girth 12, the three lines through each point in one totally isotropic plane), nine pairwise disjoint lines, and a copy of the dual polar space $DW(5,2)$ (315 lines, seven per point, collinearity graph distance-regular with intersection array $\{14,12,8;1,3,7\}$), in agreement with [M1, S, M2, LGG].

By Theorem 3, the optimal assignments of $W(5,2)$ are exactly the sign patterns of Pauli-covariant SIC fiducials in dimension 8. Our enumeration finds 120 optimal violated sets in the orbit of the Hoggar pattern under $Sp(6,2)$, and the context–point incidence matrix has rank 56 over $\mathbb{F}_2$, so each set comes from $2^7 = 128$ assignments: $120 \cdot 128 = 15360 = 240 \cdot 64$ assignments, matching the 240 Hoggar-type SICs of 64 vectors each [St]. For $Q^+(7,2)$ the orbit of our violated set under the reflections of $O^+(8,2)$ has 240 elements, the number proved in [LGG].

Proposition 3 (violated contexts in the quantum metric). Let ψ be a state, $p_v = \langle P_v \rangle$, and let $Q_{uv} = \frac{1}{4}\big(\langle P_u P_v \rangle - p_u p_v\big)$ be its quantum geometric tensor along the flows $e^{-itP_v/2}$. Consider a context $\{u,v,w\}$ with sign s and assume $p_u p_v p_w \neq 0$, so the sign assignment on this context is defined. The Fubini-Study metric is $g_{uv} = \frac{1}{4}(s\,p_w - p_u p_v)$. If $f = \operatorname{sign}(p)$, the context is violated exactly when $4|g_{uv}| = |p_w| + |p_u p_v|$ and satisfied exactly when $4|g_{uv}| = \big||p_w| - |p_u p_v|\big|$.

Proof. $\langle P_u P_v \rangle = s\,p_w$; the context is violated when $s\,p_w$ and $p_u p_v$ have opposite signs. ∎

For ψ_H: $g_{vv} = 2/9$; $|g_{uv}| = 1/9$ on the 189 commuting pairs lying on the 63 violated contexts, $1/18$ on the 756 pairs lying on satisfied contexts, $1/36$ on the 1008 anticommuting pairs; the Berry curvature has modulus $1/6$ on anticommuting pairs and vanishes on commuting ones. The hexagon is the set of pairs of largest off-diagonal metric (computed with the QGTL kernels, Section 10).

5. Saturation outside the classical families

5.1 The criterion

The statements of this section concern arbitrary sets S, which need not be closed. For an assignment f and a point $w \in S$ let the *excess* $e_w(f)$ be the number of satisfied minus the number of violated contexts of S through w. For a point $w \notin S$, $w \neq 0$, let the *outside sum* be $b_w(f) = \sum s(u,v) f(u) f(v)$ over the unordered pairs of commuting points u, v of S with $u + v = w$.

Theorem 4 (saturation criterion). Let S be a nonempty set of points, $f : S \to \{\pm 1\}$ and $A = \sum_{v \in S} f(v) P_v$. Then

$$A^2 \;=\; |S|\, I \;+\; 2 \sum_{w \in S} e_w(f)\, f(w)\, P_w \;+\; 2 \sum_{w \notin S,\; w \neq 0} b_w(f)\, P_w,$$

and the following are equivalent:

- (a) f attains the floor: $\beta_S(f) = (D-2)|S|^{3/2}/(6\sqrt{D-1})$;
- (b) A has the simple eigenvalue $\sqrt{(D-1)|S|}$ and the eigenvalue $-\sqrt{|S|/(D-1)}$ with multiplicity $D-1$;
- (c) some pure state is flat on S with signs f (Theorem 3);
- (d) $e_w(f) = (D-2)\sqrt{|S|}\,/\,(2\sqrt{D-1})$ for every $w \in S$, and $b_w(f) = 0$ for every $w \notin S$ with $w \neq 0$.

When they hold, $d(S) = \frac{1}{2}L_S - (D-2)|S|^{3/2}/(12\sqrt{D-1})$ exactly, f is optimal, and, for $N \geq 2$, $(D-1)|S|$ is a perfect square.

Proof. In $A^2 = \sum_{u,v} f(u)f(v)P_uP_v$ the diagonal terms give $|S|\,I$ and the two orders of an anticommuting pair cancel. A commuting pair $u \neq v$ contributes $2s(u,v)f(u)f(v)P_{u+v}$. If $w = u + v \in S$, the pair is a context through w and $s(u,v)f(u)f(v) = \pm f(w)$ according to whether it is satisfied; summing over the contexts through w gives $2e_w(f)f(w)P_w$. If $w \notin S$ the pairs give $2b_w(f)P_w$. This is the identity; for closed S the last sum is empty and it is Theorem 1 of [J2]. (a) $\Leftrightarrow$ (b) is the equality case of the scalar inequality, and (b) $\Leftrightarrow$ (c) is Theorem 3. Condition (b) says $(A - a)(A + a/(D-1)) = 0$ with $a = \sqrt{(D-1)|S|}$, that is $A^2 = |S|\,I + \frac{a(D-2)}{D-1}A$ (a traceless matrix with these two eigenvalues has the larger one exactly once); comparing Pauli coefficients with the identity gives (d). The floor is an integer when attained, which for $D > 2$ forces $\sqrt{(D-1)|S|}$ to be rational. ■

Condition (d) replaces the *violation-regularity* of [J2, Theorem 6], to which it reduces when every point lies on the same number of contexts. For $N \leq 5$ the number $D-1$ is squarefree, so a set that attains the floor has $|S| = (D-1)k^2$ and excess $(D-2)k/2$: $|S| \in \{3, 12\}$ for two qubits, $\{7, 28, 63\}$ for three, $\{15, 60, 135, 240\}$ for four, $\{31, 124, 279, 496, 775\}$ for five.

Table 1. The configurations we know to attain the floor (besides the stabilizer case $|S| = D-1$, $d = 0$, a maximal commuting set with the signs of a stabilizer state).

state	N	observables	contexts	degree	modulus	excess	closed	state from
$(i,i,i,1)/2$	2	12	8	0	1/2	2	no	[LLY]
CCZ applied to $(1,\ldots,1)/\sqrt{8}$	3	28	56	0	1/2	6	no	folklore
Hoggar fiducial	3	63	315	63	1/3	9	yes	[H]
$(-3,1,\ldots,1)/\sqrt{24}$	4	135	1575	315	1/3	21	yes	[J4, Ci]
ψ_B (pivot state)	4	135	1095	**75**	1/3	21	no	this work
s_5	5	496	10880	**480**	1/4	60	no	[Ci]
Φ_5 (3 CS, 2 CCZ)	5	496	10880	**480**	1/4	60	no	this work
Φ_7 (3 CS, 8 CCZ)	7	8128	2824192	**729344**	1/8	504	no	this work

The degrees in bold are new, as is the reading of the first two rows: the 12 two-qubit observables outside a maximal commuting set, and the 28 real three-qubit observables that are not diagonal, are noncontextual configurations on which the floor is nevertheless exact. The configurations of s_5 and Φ_5 are equivalent under the Clifford group (Theorem 12). The exact-degree statements for the eight masks, with their finite counts, are proved in Lean (Section 9); their state identifications are checked separately. All eight states are reproduced on Moonlab (Section 10).

5.2 A four-qubit and a five-qubit configuration

Theorem 5 (the pivot configuration). Let

$$\psi_B \;=\; \tfrac{1}{\sqrt{6}}\big(\,|0011\rangle + |0111\rangle + |1011\rangle + |1100\rangle + |1101\rangle + i\,|1110\rangle\,\big),$$

with qubit 0 written on the right, and let S_B be the set of observables with non-zero expectation value in ψ_B.

1. The expectation values are $\pm 1/3$ on S_B, and $|S_B| = 135$: the 15 diagonal observables $(0 \mid c)$, and for each $a \neq 0$ the eight observables $(a \mid c)$ with $a \cdot c = \epsilon(a)$, where $\epsilon(a) = 1$ for the five values $a \in E = \{0010, 0011, 0101, 1001, 1101\}$ and $\epsilon(a) = 0$ otherwise.
2. S_B contains 1095 contexts: 35 through each diagonal observable and 23 through each of the other 120. It is not closed and it is not a quadric, but every hyperplane $w^\perp$ meets it in 71 points if $w \in S_B$ and in 63 points otherwise, as for $Q^+(7, 2)$: it is a quasi-quadric in the sense of [DHOP]. It differs from the quadric $a \cdot c = 0$ in 80 points, those with $a \in E$, and E consists of five pairwise non-orthogonal vectors (an ovoid) of the doily $W(3, 2)$ defined by the form $x_0y_1 + x_1y_0 + x_2y_3 + x_3y_2$.
3. $d(S_B) = 75$. The assignment given by the signs of the expectation values violates the 15 lines of that doily, read among the diagonal observables, and 60 further contexts, each with exactly one diagonal observable. Every non-diagonal observable of S_B lies on exactly one violated context and every diagonal one on seven.

Proof. Items 1 and 2 and the count of 75 violated contexts are finite computations in $\mathbb{Z}[i]$ (Section 11); item 1 also follows from Proposition 6. The bound $d(S_B) \geq 75$ is Theorem 1 with $|S| = 135$, $D = 16$, $L = 1095$: $\beta \leq 14 \cdot 135 \cdot 45/(6 \cdot 15) = 945$. In Lean this is `pivot_degree`. ■

The noncontextuality inequality of this configuration has classical bound 945 and quantum value 1095 in every state. The quadric $Q^+(7, 2)$ has the same number of observables and the same classical bound, with quantum value 1575.

Proposition 6 (flat states on a bent support). Let T be the set of the six $x \in \mathbb{F}_2^4$ with $x_0x_1 + x_2x_3 = 1$, the support of a bent function, and let $\phi : T \to \mathbb{C}$ with $|\phi| = 1$. The state $\psi_\phi = 6^{-1/2} \sum_{x \in T} \phi(x)|x\rangle$ is flat on its Pauli support if and only if, up to a global phase, all $\phi(x)$ are fourth roots of unity. In that case the support has 135 points and the modulus is $1/3$; it is a hyperbolic quadric (1575 contexts, degree 315) when the number of imaginary amplitudes is even, and it has 1095 contexts and degree 75 when that number is odd. The two classes are not related by any Clifford operation.

Proof. T is a difference set with parameters $(16, 6, 2)$ [R]: every non-trivial character sum over T is ± 2, and for every $a \neq 0$ exactly one unordered pair $x_a, x_a + a$ lies in T. Hence $\langle Z^c \rangle = \pm 1/3$ for $c \neq 0$, and for $a \neq 0$, with $z_a = \overline{\phi(x_a + a)}\, \phi(x_a)$,

$$\langle X^a Z^c \rangle = \tfrac{1}{6}(-1)^{c \cdot x_a} \big(z_a + (-1)^{c \cdot a}\, \overline{z_a} \big),$$

so the expectation value of $P_{(a|c)}$ is $\pm \frac{1}{3} \operatorname{Re} z_a$ when $a \cdot c = 0$ and $\pm \frac{1}{3} \operatorname{Im} z_a$ when $a \cdot c = 1$. A common modulus must be $1/3$, which holds exactly when every z_a is ± 1 or $\pm i$, that is, when all ratios of amplitudes are fourth roots of unity. Then for each $a \neq 0$ exactly the eight observables with $a \cdot c = \epsilon(a)$ are in the support, where $\epsilon(a) = 0$ if z_a is real and 1 if it is imaginary. Write $\phi = i^k$ and $\kappa = k \bmod 2$; then $\epsilon(x + x') = \kappa_x + \kappa_{x'}$. A diagonal Clifford operation multiplies ϕ by $i^{q(x)}$ for a $\mathbb{Z}_4$-valued quadratic form q and changes κ by the restriction to T of a linear functional; these sixteen restrictions have weights 0, 2 and 4, and together with the all-ones vector (a global factor i) they span the even-weight vectors of $\mathbb{F}_2^T$. So κ of even weight is equivalent to $\kappa = 0$, a real state, whose support is the quadric $a \cdot c = 0$; and κ of odd weight is equivalent to a single imaginary amplitude, as in ψ_B. Symplectic maps preserve the number of contexts, and $1575 \neq 1095$. The two counts were computed for all 32 supports that occur, over all 4096 phase patterns (Section 11). ■

Theorem 7 (five qubits). Let S_5 be the set of the 496 five-qubit observables with non-zero expectation value in the state s_5 of Cieśliński [Ci, eq. (18)], all equal to $\pm 1/4$. Then S_5 contains 10880 contexts and $d(S_5) = 480$. The optimal assignment, the signs of the expectation values, leaves 16 observables on no violated context (60 contexts through each), 240 on two (of 64) and 240 on four (of 68). Every hyperplane $w^\perp$ meets S_5 in 256 points if $w \in S_5$ and in 240 otherwise, as for the complement of $Q^+(9,2)$; S_5 is not such a complement.

Proof. The counts are finite computations in $\mathbb{Z}[i]$. The lower bound is Theorem 1 with $|S| = 496$, $D = 32$: $\beta \le 30{\cdot}496{\cdot}124/(6{\cdot}31) = 9920 = 10880 - 2{\cdot}480$. In Lean this is `cieslinski5_degree`. No state vanishes on every observable of a hyperbolic quadric: these are the observables commuting with an antiunitary θ with $\theta^2 = 1$, and the condition would give $\rho + \theta\rho\,\theta^{-1} = 2I/D$, of rank D, for a projector ρ of rank one. ∎

Proposition 8 (the optimum in the quantum metric). Let ψ be flat on S with modulus μ, and let f be the signs of its expectation values. For $u \ne v$ in S the Fubini–Study metric of Proposition 3 has $|g_{uv}| = (\mu + \mu^2)/4$ if $u, v, u+v$ form a context of S violated by f, $|g_{uv}| = (\mu - \mu^2)/4$ if they form a satisfied context of S, and $|g_{uv}| = \mu^2/4$ otherwise (u, v anticommute, or their product is outside S). So the violated contexts of the optimal assignment are exactly the pairs with $4|g_{uv}|$ larger than μ. The Berry curvature has modulus $\mu/2$ on anticommuting pairs whose product is in S and vanishes on all other pairs.

Proof. Proposition 3 with $|p| = \mu$ on S and $p = 0$ outside; for anticommuting u, v the product $P_u P_v$ is $\pm i P_{u+v}$, so $\langle P_u P_v \rangle$ is imaginary, of modulus $|p_{u+v}|$. ∎

Proposition 9 (two mutually unbiased stabilizer states). Let s_1, s_2 be stabilizer state vectors whose underlying Pauli label spaces are complementary: they intersect only in the zero label. Then $|\langle s_1|s_2\rangle| = D^{-1/2}$, and let $\varphi = \theta + \arg\langle s_1|s_2\rangle$. The normalized state proportional to $s_1 + e^{i\theta} s_2$ is flat on a set that is not a stabilizer group exactly in the following cases: $N = 1$ with $\cos\varphi = (\sqrt{6} - \sqrt{2})/4$ or $-(\sqrt{6} + \sqrt{2})/4$ (all 3 observables, modulus $1/\sqrt{3}$); $N = 2$ with $\varphi = \pm\pi/2$ (12 observables, modulus $1/2$); $N = 3$ with $\varphi = \pm 3\pi/4$ (all 63, modulus $1/3$: the Hoggar fiducial is $\sqrt{8}$ times the uniform vector plus $(-2+2i)$ times the first basis vector); $N = 4$ with $\varphi = \pi$ (the 135 observables of a hyperbolic quadric, modulus $1/3$: $(-3, 1, \ldots, 1)$ is 4 times the uniform unit vector minus 4 times the first basis vector). There is none for $N \ge 5$.

Proof. Let $t = D^{-1/2}$ and $n = 2 + 2t\cos\varphi$ the squared norm. An observable of the first stabilizer group has expectation $\pm(1 + 2t\cos\varphi)/n$, and so has one of the second. Every other observable is, up to a phase, a product $P_1 P_2$ of non-identity elements of the two groups, and its expectation is $\pm 2t\cos\varphi/n$ if P_1, P_2 commute and $\pm 2t\sin\varphi/n$ if they anticommute. Equating the non-zero moduli: all three are equal only if $|\cos\varphi| = |\sin\varphi|$ and $1 - \sqrt{2}\,t = \sqrt{2}\,t$, so $D = 8$; with $\sin\varphi = 0$ one needs $1 - 2t = 2t$, so $D = 16$ and $\cos\varphi = -1$; with $\cos\varphi = 0$ one needs $1 = 2t$, so $D = 4$; for $D = 2$ there are no commuting products and $|1 + \sqrt{2}\cos\varphi| = \sqrt{2}|\sin\varphi|$ has the two stated solutions. The remaining case $1 + 2t\cos\varphi = 0$ gives a stabilizer state. ∎

So the SIC fiducials of one and three qubits, a state with the Pauli spectrum of [LLY], and the quadric state are four members of one family, and that family stops at four qubits. The pivot state is not in it: it is a combination of three pairwise unbiased stabilizer states.

5.3 Flat phase states, and a seven-qubit configuration

Proposition 10 (the support is a two-character set). Let ψ be flat on S and let $k^2 = |S|/(D-1)$, the inverse square of the modulus. For every point w,

$$\sum_{v \in S} (-1)^{\langle v,w\rangle} \;=\; D\,[w \in S] \;-\; k^2.$$

Equivalently, exactly $(D-2)|S|/(2(D-1)) + \frac{D}{2}[w \in S]$ observables of S commute with P_w. Hence the graph on $\mathbb{F}_2^{2N}$ in which u and v are adjacent when $u+v \in S$ is strongly regular (possibly imprimitive, or complete), with degree $|S|$ and nonprincipal character eigenvalues among $D-k^2$ and $-k^2$ [CK]; an empty character class contributes no eigenvalue. If $k^2 = D/2$ the sum takes the values $\pm D/2$: then S is a Hadamard difference set, the support of a bent function on the $2N$-dimensional space.

Proof. For a pure state ρ and any w, $\mathrm{tr}(\rho P_w \rho P_w) = \langle P_w \rangle^2$. Writing $\rho = D^{-1}\sum_v p_v P_v$ with $p_0 = 1$ and using $P_w P_v P_w = (-1)^{\langle v,w\rangle} P_v$ gives $p_w^2 = D^{-1}\sum_v (-1)^{\langle v,w\rangle} p_v^2$. Insert $p_v^2 = k^{-2}$ on S, 1 at $v=0$ and 0 elsewhere. ∎

The identity for p_w^2 holds for every pure state and is standard; we did not find its consequence for flat states stated. It accounts for the hyperplane numbers in Theorems 5 and 7. For the two four-qubit rows of Table 1 it gives a strongly regular graph with the parameters $(256, 135, 70, 72)$ of the affine polar graph of $Q^+(7,2)$; for the three-, five- and seven-qubit rows with modulus $2^{-(N-1)/2}$ it gives bent functions in 6, 10 and 14 variables.

Phase states. Let $a_j \in \mathbb{Z}_8$, $b_{jk} \in \mathbb{Z}_4$, $c_{jkl} \in \mathbb{Z}_2$ and

$$p(x) = \sum_j a_j x_j + 2\sum_{j<k} b_{jk}\, x_j x_k + 4 \sum_{j<k<l} c_{jkl}\, x_j x_k x_l \pmod 8, \qquad |\Phi_p\rangle = D^{-1/2} \sum_x e^{i\pi p(x)/4} |x\rangle.$$

Φ_p is obtained from $|+\cdots+\rangle$ by T gates (a_j), controlled-S gates CS (b_{jk}) and CCZ gates (c_{jkl}); these are the diagonal gates of the third level of the Clifford hierarchy [CGK]. Let T be the symmetric trilinear form on $\mathbb{F}_2^N$ with $T(e_j,e_j,e_j) = a_j$, $T(e_j,e_j,e_k) = T(e_j,e_k,e_k) = b_{jk}$ and $T(e_j,e_k,e_l) = c_{jkl}$ modulo 2. Then $B(x,z) = T(x,x,z)$ is a symmetric bilinear form, alternating exactly when there is no T gate. A linear change of variables in T is a CNOT circuit on the state, and two polynomials with the same form differ by a diagonal Clifford gate.

Proposition 11 (phase states of the third level). For $a \neq 0$ let R_a be the radical and r_a the rank of the symmetric bilinear form $T(a,\cdot,\cdot)$. The expectation value of $X^a Z^c$ in Φ_p has modulus $2^{-r_a/2}$ for the 2^{r_a} vectors c of one coset of $R_a^\perp$ and vanishes for every other c; diagonal observables have expectation value zero. So Φ_p has a single non-zero modulus exactly when $r_a = r$ is the same for every $a \neq 0$. Then the modulus is $2^{-r/2}$, the support has $(D-1)2^r$ points, r is even if $N \geq 2$, and Φ_p attains the floor on its support. If moreover $r = N-1$, each R_a has one non-zero element $\rho(a)$, the map ρ is an involution of the non-zero vectors, and

$$S = \{\, (a \mid c) \;:\; a \neq 0, \quad c \cdot \rho(a) = \epsilon(a) \,\}$$

for a function ϵ: the support of a bent function of Maiorana–McFarland type [McF].

Proof. Modulo 8 the third difference of p is $4T$, independent of the base point, and the second difference is even. So $p(x) - p(x+a) = h_a + 2q_a(x)$ with $q_a(0) = 0$ and $q_a(x+y) = q_a(x) + q_a(y) + 2T(a,x,y)$ modulo 4, a $\mathbb{Z}_4$-valued quadratic form [CCKS], and $\langle X^a Z^c \rangle = D^{-1} e^{i\pi h_a/4} \sum_x i^{q_a(x)} (-1)^{c\cdot x}$. Now

$$\Big| \sum_x i^{q_a(x)} (-1)^{c\cdot x} \Big|^2 = \sum_z i^{q_a(z)} (-1)^{c\cdot z} \sum_y (-1)^{T(a,y,z)} = D \sum_{z \in R_a} i^{q_a(z)} (-1)^{c\cdot z}.$$

On R_a the map $z \mapsto i^{q_a(z)}$ is a character with values ± 1, so the last sum is 2^{N-r_a} when $(-1)^{c\cdot z}$ agrees with it on R_a and 0 otherwise. That r is even follows from Theorem 4. If $r = N-1$ then $T(a,\rho(a),\cdot) = 0$, which is symmetric in a and $\rho(a)$, so $\rho(\rho(a)) = a$. ∎

Theorem 12 (all flat phase states of the third level on at most five qubits). Among the non-zero forms T in $N \le 5$ variables, those of constant rank are the following.

N	forms	of constant rank	classes under CNOT circuits	gates of a representative	r	observables
1	1	1	1	T	1	2
2	7	1	1	$CS(0,1)$	2	12
3	127	8	2	$CCZ(0,1,2)$; $CS(0,1)\ CS(0,2)$	2	28
4	16383	0				
5	33554431	166656	1	$CS(0,1)\ CS(1,3)\ CS(2,4)$ $CCZ(0,1,4)\ CCZ(0,2,3)$	4	496

For five qubits the stabilizer of the form in $GL(5,2)$ has order 60. Write Φ_5 for the state of the last row. The supports of the two three-qubit classes are equivalent under $Sp(6,2)$, and the support of Φ_5 is mapped onto the set S_5 of Theorem 7 by a symplectic map of $\mathbb{F}_2^{10}$. Thus no state with nonzero associated form T in this family is flat on four qubits; on five qubits every flat state with nonzero T is obtained from Φ_5 by a CNOT circuit and a diagonal Clifford gate; and the configuration found numerically in [Ci] is, up to a Clifford operation, the Pauli support of a circuit of five gates. The excluded zero form gives diagonal-Clifford stabilizer states, which are flat in every number of qubits.

Proof. Exhaustive enumeration, with the orbits computed from the transvections that generate $GL(N,2)$ (`src/third_level.c`; for $N \le 4$ also `scripts/phase_states.py`). The two symplectic maps are found by search and verified on every point; the second is recorded in `results/phase_states.json`. ∎

For $N = 2$ the state is that of [LLY] up to a phase and complex conjugation. In coordinates $X = (x_1, x_2, \alpha) \in \mathbb{F}_q \times \mathbb{F}_q \times \mathbb{F}_2$ with $q = 4$, the five-qubit form is

$$T(X,Y,Z) = \mathrm{Tr}\left[x_1y_1z_2^2 + x_1y_2^2z_1 + x_2^2y_1z_1 + x_2y_2z_1^2 + x_2y_1^2z_2 + x_1^2y_2z_2\right] + \alpha\,\sigma(y,z) + \beta\,\sigma(x,z) + \gamma\,\sigma(x,y),$$

where $\sigma(y,z) = \mathrm{Tr}[a\,(y_1z_2 + y_2z_1)]$, Tr is the trace to $\mathbb{F}_2$ and a generates $\mathbb{F}_4$. It is invariant under $(x_1, x_2) \mapsto (\lambda x_1, \lambda^{-1}x_2)$.

Theorem 13 (seven qubits). Let

$$|\Phi_7\rangle = CS_{01}\,CS_{23}\,CS_{45}\ \ CCZ_{012}\,CCZ_{035}\,CCZ_{124}\ \ CCZ_{046}\,CCZ_{056}\,CCZ_{136}\,CCZ_{236}\,CCZ_{256}\ |+\rangle^{\otimes 7}$$

and let S_7 be its Pauli support.

1. Φ_7 has expectation values $\pm 1/8$ on S_7 and 0 elsewhere, and $|S_7| = 8128 = 2^6(2^7 - 1)$.
2. S_7 contains 2 824 192 contexts and $d(S_7) = 729\,344$. In the optimal assignment every observable of S_7 lies on 504 more satisfied than violated contexts; the number of contexts through an observable is 1008, 1024, 1040, 1056, 1072 or 1104 (for 64, 3584, 1792, 1344, 896 and 448 observables).
3. Let $\mathcal{F}$ be the family of forms on $\mathbb{F}_2^7$ for which B is alternating of rank 6. The forms of constant rank in $\mathcal{F}$ are a single class under CNOT circuits, and the stabilizer of such a form in $GL(7,2)$ has order 7. In coordinates $\mathbb{F}_8 \times \mathbb{F}_8 \times \mathbb{F}_2$ the form is the one displayed above, with the trace of $\mathbb{F}_8$, with $a^3 + a^2 + 1 = 0$, and with the additional term $\mathrm{Det}(x_1, y_1, z_1)$, the alternating trilinear form of $\mathbb{F}_8$ over $\mathbb{F}_2$.

4. For each of the ten other types of B (alternating of rank 0, 2 or 4; not alternating, of rank 1 to 7) we tested 3×10^8 random forms; none has constant rank.

Proof. Item 1 and the counts of item 2 are finite computations in $\mathbb{Z}[i]$, made exactly by `scripts/phase_states.py`, which records the counts per observable in `results/phase_states.json`. Two programs that share no code with it recount the totals 8128, 2 824 192 and 729 344 (Section 11): the Moonlab program from the expectation values of the simulated state, which also give the support and the modulus 1/8 up to rounding, and the QGTL program in C from the support and signs that `scripts/phase_states.py` writes to `data/flat/`. The lower bound is Theorem 1 with $|S| = 8128$, $D = 128$: $\beta \le 126 \cdot 8128 \cdot 1016/(6 \cdot 127) = 1\,365\,504 = 2\,824\,192 - 2 \cdot 729\,344$. In Lean, for the set of observables given by the mask of S_7, these are `phase7_contexts` and `phase7_degree` (Section 9.1). For item 3, choose a basis in which e_6 spans the radical of B and $e_0, \dots, e_5$ is a symplectic basis of a complement. Every form of $\mathcal{F}$ is then the third difference of $2(x_0x_1 + x_2x_3 + x_4x_5) + 4w(x_0, \dots, x_5) + 4x_6\, s(x_0, \dots, x_5)$ for a set w of CCZ gates on the first six qubits (2^{20} choices) and a graph s on them, which must be nondegenerate (13888 choices). All 1.46×10^{10} such forms were tested (`src/family7.c`): 13 271 040 have constant rank. The group that preserves the normal form has order $64\,|Sp(6,2)| = 92\,897\,280$, and the solutions are one orbit of it, of size $92\,897\,280/7$ (`src/orbits7.c`). ∎

The noncontextuality inequality of S_7 has classical bound 1 365 504 and quantum value 2 824 192: no noncontextual model reproduces more than 74.2 per cent of its contexts. Item 4 is evidence, not proof, that Φ_7 represents the only class of flat phase states with nonzero associated form on seven qubits, up to CNOT circuits and diagonal Clifford gates. The numbers of observables for three, five and seven qubits, 28, 496 and 8128, are $2^{N-1}(2^N - 1)$; that they are the second, third and fourth perfect numbers reflects only that 7, 31 and 127 are prime. We do not know whether this family extends to a nine-qubit flat phase state of constant rank eight, which would have 130816 observables of modulus 1/16. Proposition 11 does not rule out other constant even ranks. The tested forms with the symmetry $(x_1, x_2) \mapsto (\lambda x_1, \lambda^{-1} x_2)$ of the five- and seven-qubit ones do not have constant rank for nine or eleven qubits.

5.4 What does not attain the floor

For the $D^2 - D$ observables outside a maximal commuting set (the non-diagonal ones, say) there are $(D-1)(D-2)D^2/12$ contexts and Theorem 1 gives a violated fraction of at least $\frac{1}{2} - 2^{-N/2}$. By Theorem 4, equality would need a state with all non-diagonal expectation values of modulus $2^{-N/2}$, that is, a state whose orbit under the Pauli group forms, with the computational basis, a complete set of mutually unbiased bases. For two qubits this is the first row of Table 1. For four qubits and for every odd $N \ge 3$ no such state exists [K], so the degree is strictly above the floor there. A numerical search from random starts found flat four-qubit states only with $|S| = 135$ (both classes of Proposition 6) and flat five-qubit states only with $|S| = 496$; it found none with $|S| = 60$, or with $|S|$ equal to 124, 279 or 775. This is evidence, not proof, that Table 1 is complete up to five qubits.

6. Consequences for all N

Corollary A (full space). For every $N \ge 1$ every assignment of the $D^2 - 1$ Pauli observables violates at least $\varphi(D)\, L$ contexts (Lean: `full_space_fraction`). Since a uniformly random assignment violates $L/2$ contexts on average, $d/L \to 1/2$.

The count inequality also covers $N = 1$, when there are no contexts. The ratios and noise thresholds below are stated for $N \ge 2$, so $L > 0$.

Exponential violation. The ratio of the quantum value to the noncontextual bound of the full inequality satisfies $L/(L-2d) \geq (D+2)/(2\sqrt{D+1}) \sim 2^{N/2-1}$. Cabello [C] proposed this inequality in 2010 and argued for exponential growth; his Lemma 3 identifies the bound with the number of positive contexts, which is correct for two qubits but not for three, where 252 of the 315 predictions can be satisfied rather than 225 [M1]. Corollary A proves the growth.

Noise. If imperfections scale every correlator by a visibility η, the inequality is violated when $\eta > 1 - 2d/L$, hence whenever $\eta > 2\sqrt{D+1}/(D+2)$: above 0.6 for three qubits, 0.458 for four, 0.338 for five, 0.062 for ten. In the notation of [C, MS] the tolerated error per context $\varepsilon = 2d/L$ is at least $1 - 2\sqrt{D+1}/(D+2)$, which exceeds 0.707 (the largest value confirmed in the search of [MS], for configurations of a different kind) from six qubits on and tends to 1.

Proposition 4 (the floor is sharp up to a logarithmic factor). Let $N \geq 2$. For every pure N-qubit state with $M = \max_{v \neq 0} |\langle P_v \rangle|$ there is an assignment of the full space with

$$\beta \;\geq\; \frac{(D-1)(D-2)}{6\,M^3}, \qquad \text{so} \qquad \frac{1}{(D+1)(D+2)\,M_N^3} \;\leq\; \frac{1}{2} - \frac{d}{L} \;\leq\; \frac{\sqrt{D+1}}{D+2},$$

where M_N is the least value of M over all states. $M_N \geq 1/\sqrt{D+1}$, with equality exactly when a Pauli-covariant SIC exists, and $M_N \leq C\sqrt{\log D / D}$ for an absolute constant C.

Proof. A pure state has $\operatorname{tr}\rho^3 = 1$; with $\rho = D^{-1}(I + \sum_v p_v P_v)$ and $\sum_{v \neq 0} p_v^2 = D - 1$, the trace computation of Theorem 1 gives $T(p) = \sum_\ell s_\ell\, p_u p_v p_w = (D-1)(D-2)/6$ (the cubic purity condition [JL, St]). Choose the signs $f(v)$ independently with mean p_v/M. Then $\mathbb{E}\,\beta(f) = T(p)/M^3$, and some f does at least as well. The upper bound on M_N follows from Lévy's lemma for Haar-random states and a union bound over the $D^2 - 1$ observables [HLW]. ∎

So $\frac{1}{2} - d/L = \Theta(2^{-N/2})$ up to a factor $\log^{3/2} D$. In four sample runs the sign patterns of Haar-random states reached between 50 and 64 per cent of the floor value of β for four and five qubits. We did not find Proposition 4 in the notes we could read; we do not claim it is new.

Averaging between families. Every context of $W(7,2)$ lies on exactly 24 of its 120 elliptic quadrics, and every context of $Q^+(9,2)$ lies in exactly 112 of its 496 sections by an imaginary Pauli observable, each a signed copy of $W(7,2)$ [LGG, J1]; we re-verified these incidences by enumeration. With Theorem 2 and integer rounding this gives $d(Q^-(7,2)) \geq 291$, $d(W(7,2)) \geq 1455$, $d(Q^+(9,2)) \geq 6444$, $d(W(9,2)) \geq 28791$, each below the values of [J1, J4] and stated here only to show what the floor alone yields.

7. The data against the floor

Table 2 confronts every published or pre-printed value we know for the classical families with the floor. "Floor" is the integer bound of Theorem 1 (Theorem 2 for elliptic quadrics), computed in exact integer arithmetic. "Best assignment" is the smallest number of violated contexts known; all entries up to five qubits were reproduced by our own search (Section 11).

Table 2. Classical configurations compared with the spectral floor.

configuration	N	points	contexts	averaging bound $\lceil L/5 \rceil$ [M2, LGG]	floor	best assignment	exact value
$W(3,2)$	2	15	15	3	2	3	3 [dB]
$Q^+(3,2)$	2	9	6	–	1	1	1
$W(5,2)$	3	63	315	63	**63**	63	63 [M1]
$Q^+(5,2)$	3	35	105	21	14	21	21 [dB]
$Q^-(5,2)$	3	27	45	9	**9**	9	9 [dB]

configuration	N	points	contexts	averaging bound $\lceil L/5 \rceil$ [M2, LGG]	floor	best assignment	exact value
$W(7,2)$	4	255	5355	1071	1451	1575 [M2]	1575 [J1]*
$Q^+(7,2)$	4	135	1575	315	**315**	315 [M2]	315 [LGG]
$Q^-(7,2)$	4	119	1071	215	291	315 [M2]	315 [J1]*
$W(9,2)$	5	1023	86955	17391	28786	31479 [M2]	open ($\geq$ 28830 [J4]*)
$Q^+(9,2)$	5	527	23715	4743	6426	6975 [M2]	6975 [J1]*
$Q^-(9,2)$	5	495	19635	3927	6501	7087 [M2]	open ($\geq$ 6510 [J4]*)
$W(11,2)$	6	4095	1396395	279279	527621	553140 [M2]	open
$Q^+(11,2)$	6	2079	365211	73043	120901	132391 [M2]	open
$Q^-(11,2)$	6	2015	332475	66495	125624	131700 [M2]	open
$W(13,2)$	7	16383	22362795	4472559	9227609	9406024 [M2]	open
$Q^+(13,2)$	7	8255	5720715	1144143	2161542	2331191 [M2]	open
$Q^-(13,2)$	7	8127	5458635	1091727	2252409	2294580 [M2]	open

* computer-assisted, not formally verified, not peer reviewed; we did not re-run these computations.

Reading of the table. (1) All 17 rows are consistent with the floor; it is attained in the three rows in bold and, after rounding, in $Q^+(3,2)$, the Mermin–Peres square. (2) From four qubits on (five for hyperbolic quadrics) the floor exceeds the averaging bound $L/5$, and the gap to the best assignment shrinks: the floor closes 75, 81, 91 and 96 per cent of the interval between $L/5$ and the best assignment for the full space at $N = 4, 5, 6, 7$. (3) For seven qubits the best known assignments of the full space and of the elliptic quadric are within two per cent of the floor. (4) Outside these families the floor is exact for the configurations of Table 1.

An empirical regularity, also recorded for four qubits in [J1]: in every best assignment we computed, $A + 3I$ has low rank. The spectra of A are $\{21 \pm 16\sqrt{6}, (-3)^{14}\}$ for $W(7,2)$, $\{(9 \pm 4\sqrt{23})^2, (-3)^{12}\}$ for $Q^-(7,2)$, $\{45 \pm 56\sqrt{2}, (-3)^{30}\}$ for $Q^+(9,2)$, and three (respectively three pairs of) eigenvalues different from -3 for $W(9,2)$ (respectively $Q^-(9,2)$).

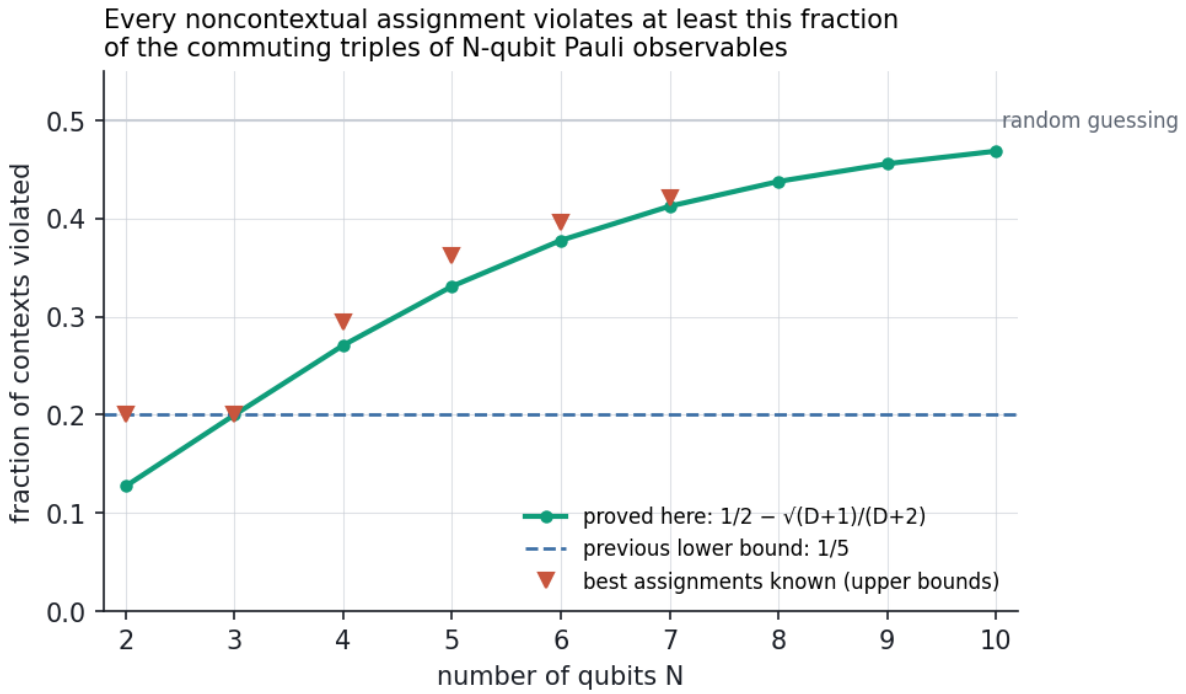


Figure 1. Lower bound on the fraction of violated contexts for the full N-qubit Pauli configuration, the earlier bound 1/5, and the best assignments known.

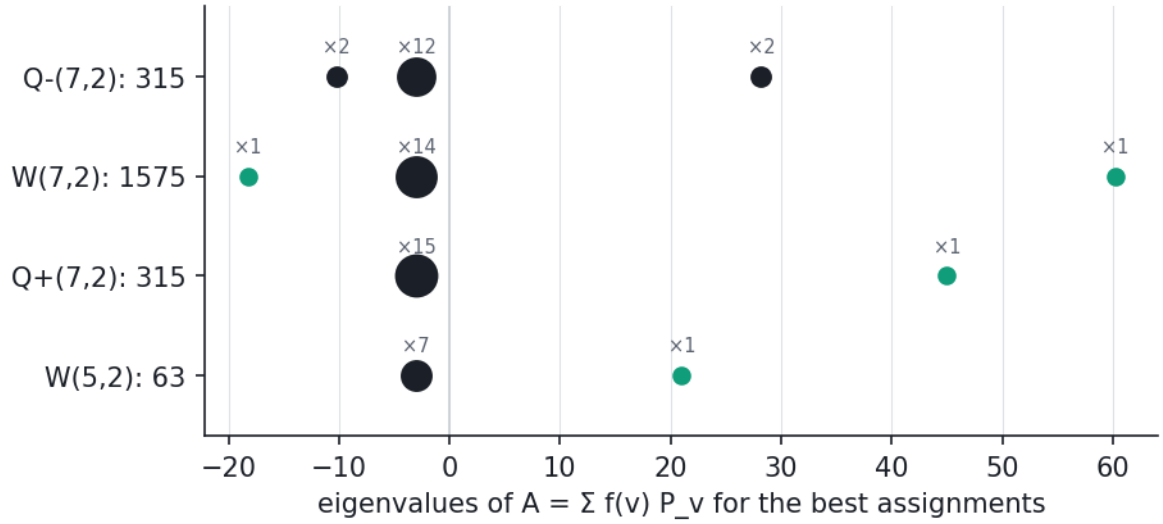


Figure 2. Spectra of $A = \sum f(v)P_v$ for the best assignments at three and four qubits.

8. What a carrier spectrum does not determine

8.1 The completion question

For a fixed assignment on a hyperplane section, the remaining contextuality constraints define a quadratic problem. Jonah's *strict law* says that equal spectra of the section carriers imply equal spectra of the corresponding completion operators within a specified class [J3]. Sign carriers, whose coefficients all belong to $\{-1, +1\}$, impose a stronger admissibility requirement than general real carriers: a construction using zero coefficients need not answer the sign-carrier question.

[J3, Theorem B4] lists sixteen hyperplane cases. Nine of them have a section with a vertex: C1, C3, C5, C6, C8, C11, C12, C14 and C15. For real carriers the strict law fails in all nine [J3, Theorem B2]; for sign carriers [J3] proves failure in C1 and C6 and states:

> **Conjecture E** [J3, Section 6]. On sign carriers the strict law fails in every vertex case for every $N \geq 4$.

Case C3 is a hyperbolic quadric cut by a hyperplane whose polar point lies on the quadric; the inspected source leaves it open for $N \geq 4$. We first give two admissible carriers at $N = 4$ and prove the predicted failure there (Sections 8.2 to 8.5). Section 8.6 gives a family for every $N \geq 4$, and Section 8.7 constructions for all nine vertex cases, which prove the conjecture. This concerns the information in the entire carrier spectrum, even when its eigenvalues are known exactly; it leaves the spectral floor of Section 3 valid. Case names C1 to C15 are those of [J3]; the equation tags (C1) to (C12) refer to the four-qubit certificate, and (E1) to (E12) to Sections 8.6 and 8.7.

8.2 Geometry and two complete witnesses

Set $N = 4$, $D = 16$, and let $\mathcal{Q} = Q^+(7, 2)$ be the 135 nonidentity real Pauli observables, equivalently the labels with $q_0(a, c) = a \cdot c = 0$ modulo two. Use the polar point $p_\star = ZIII$ and write

$$H = \{v \in \mathcal{Q} : \langle v, p_\star \rangle = 0\}, \qquad O = \mathcal{Q} \setminus H.$$

Here H is the section and O its complement; the letter S retains its general meaning from Section 2. If R is the set of 35 real nonidentity three-qubit Paulis, then

$$H = \{Z \otimes I_8\} \cup \{I \otimes P, Z \otimes P : P \in R\}.$$

Thus $|H| = 71$ and $|O| = 64$. The outside points have first factor X with a real three-qubit factor, or first factor Y with an imaginary three-qubit factor: 36 and 28 points respectively. Each line not contained in H contains exactly one section point and two outside points. The quadric has 1575 contexts, of which 1120 are outside lines; these numbers refer to $\mathcal{Q}$, not S_B.

Define $\sigma_0, \sigma_1 : H \to \{-1, +1\}$ to be positive except at these labels:

Table 3. Complete C3 sign witnesses.

Carrier	Negative Pauli words	Negative numeric labels
σ_0	IIZZ, IYYI, ZIZZ, ZYYI	48, 102, 176, 230
σ_1	IYXY, ZIIZ, ZIZZ, ZYXY	87, 144, 176, 215

Both give $+1$ to $p_\star$. No search seed or external assignment file is needed to specify them. Their carrier operators are $A_\sigma = \sum_{s \in H} \sigma(s) P_s$. Define the real symmetric completion matrix Λ_σ, indexed by O, by

$$(\Lambda_\sigma)_{uv} = (\Lambda_\sigma)_{vu} = s_\ell \sigma(s) \quad \text{when } \ell = \{s, u, v\},\ s \in H,\ u, v \in O,$$

and zero for all other entries, including the diagonal. For $y \in \{-1, +1\}^O$ the number of violated outside lines is

$$F(\sigma, y) = 560 - \frac{1}{4} y^{\mathsf{T}} \Lambda_\sigma y. \tag{C1}$$

Each line contributes $(1 - s_\ell \sigma(s) y_u y_v)/2$; the symmetric quadratic form counts it twice. This proves the normalization and agrees with [J3, Section 2].

8.3 Exact statement

Theorem 14 (four-qubit C3 certificate). The witnesses in Table 3 have equal characteristic polynomials,

$$\begin{aligned} \chi_{A_{\sigma_0}}(t) = \chi_{A_{\sigma_1}}(t) = (t-3)^2(t+1)^8(t+5)^2(t+13) \\ \cdot (t^3 - 25t^2 - 133t + 1621), \end{aligned} \tag{C2}$$

but their completion operators satisfy

$$\operatorname{tr} \Lambda_{\sigma_0}^3 = 14976, \qquad \operatorname{tr} \Lambda_{\sigma_1}^3 = 15360. \tag{C3}$$

Consequently the carrier spectra, including multiplicities, are equal and the completion spectra are different. This proves failure of the strict law for four-qubit C3 sign carriers, establishing the corresponding instance of Conjecture E.

8.4 Block proof and the information that is lost

Every section operator commutes with $p_\star$, so in its ± 1 eigenspaces

$$A_\sigma = \operatorname{diag}(B_{\sigma,+}, B_{\sigma,-}).$$

Put $\alpha = \sigma(Z \otimes I)$, $a_P = \sigma(I \otimes P)$ and $b_P = \sigma(Z \otimes P)$. The two real symmetric 8×8 blocks are

$$B_{\sigma,+} = \alpha I + \sum_{P \in R} (a_P + b_P) P, \qquad B_{\sigma,-} = -\alpha I + \sum_{P \in R} (a_P - b_P) P. \tag{C4}$$

Since $a_P, b_P \in \{-1, +1\}$, each P occurs in exactly one block with coefficient $+2$ or -2. For three-qubit codes use $a + 8c$ and let $T_R = \sum_{P \in R} P$. Table 3 gives

$$\begin{aligned} B_{0,+} &= I + 2(T_R - 2P_{24} - 2P_{54}), & B_{0,-} &= -I, \\ B_{1,+} &= I + 2(T_R - P_8 - P_{24} - 2P_{47}), & B_{1,-} &= -I + 2P_8 + 2P_{24}. \end{aligned} \tag{C5}$$

Here $P_8 = IIZ$, $P_{24} = IZZ$, $P_{47} = YXY$ and $P_{54} = YYI$. Define

$$g(t) = t^3 - 25t^2 - 133t + 1621, \qquad h(t) = (t+13)g(t).$$

Exact determinant evaluation in (C5) yields the following finite integer identities.

Table 4. The block characteristic polynomials.

Block	Characteristic polynomial
$B_{0,+}$	$(t-3)^2(t+5)^2 h(t)$

Block	Characteristic polynomial
$B_{0,-}$	$(t+1)^8$
$B_{1,+}$	$(t+1)^4 h(t)$
$B_{1,-}$	$(t-3)^2(t+1)^4(t+5)^2$

They can be reproduced directly from the explicitly specified eight-dimensional matrices; no approximate root comparison or classification theorem is involved. The last row also follows from the joint eigenvalues of its two commuting Z-type Paulis. Multiplying the plus and minus polynomials in each pair proves (C2).

The unlabelled union of block spectra stays fixed while their allocation changes. Two copies each of 3 and -5 move from the plus block to the minus block, and four copies of -1 move in the reverse direction. The common carrier spectrum alone does not record this allocation.

The outside Pauli span is the space of real symmetric off-diagonal block matrices

$$\begin{pmatrix} 0 & M \\ M^{\mathsf{T}} & 0 \end{pmatrix}, \qquad M \in \mathbb{R}^{8\times 8}.$$

One half of the anticommutator with A_σ acts there as

$$M \longmapsto \frac{1}{2}(B_{\sigma,+}M + MB_{\sigma,-}). \tag{C6}$$

Its matrix in the outside Pauli basis is exactly Λ_σ: commuting Pauli pairs supply the context sign and anticommuting pairs vanish. The polar term contributes zero on this outside span. This is the C3 Jordan representation of [J3, Theorem A and Table 1]. In orthonormal block eigenbases,

$$\operatorname{spec}\Lambda_\sigma = \left\{ \frac{b_{+,i} + b_{-,j}}{2} : 1 \le i, j \le 8 \right\}, \tag{C7}$$

where each ordered pair contributes once. These sums depend on the block allocation that the carrier spectrum has discarded. Expanding their cubes gives

$$\begin{aligned} \operatorname{tr}\Lambda_\sigma^3 = {} & \operatorname{tr} A_\sigma^3 \\ & + \frac{3}{8}\big[(\operatorname{tr} B_{\sigma,+}^2)(\operatorname{tr} B_{\sigma,-}) + (\operatorname{tr} B_{\sigma,+})(\operatorname{tr} B_{\sigma,-}^2)\big]. \end{aligned} \tag{C8}$$

For both witnesses $\operatorname{tr} B_+ = 8$, $\operatorname{tr} B_- = -8$ and $\operatorname{tr} A^3 = 18336$. If r real nonidentity three-qubit Paulis occur in the plus block, Pauli orthogonality gives

$$\operatorname{tr} B_+^2 = 8(1+4r), \qquad \operatorname{tr} B_-^2 = 8(1+4(35-r)). \tag{C9}$$

For σ_0, $r = 35$, so these traces are 1128 and 8; for σ_1, $r = 33$, so they are 1064 and 72. Equation (C8) therefore gives

$$18336 + 3(8 - 1128) = 14976, \qquad 18336 + 3(72 - 1064) = 15360.$$

Equal spectra would have equal cubic traces. Their difference of 384 proves (C3) and the spectral conclusion without numerical diagonalization. ■

8.5 Consequences for the completion optima

Corollary 15. The largest completion eigenvalues differ by exactly two:

$$\lambda_{\max}(\Lambda_{\sigma_1}) - \lambda_{\max}(\Lambda_{\sigma_0}) = 2. \tag{C10}$$

Let ρ be the largest root of g. The values $g(25) = -1704$, $g(30) = 2131$ and $g'(t) = 3t^2 - 50t - 133 > 0$ for $t \geq 25$ show that exactly one root lies above 25, and it belongs to $(25, 30)$. All roots are real because they occur in a real symmetric block polynomial. Both plus blocks have largest eigenvalue ρ; their other explicit eigenvalues are at most 3. The minus-block maxima are -1 and 3 respectively. Equation (C7) gives $(\rho - 1)/2$ and $(\rho + 3)/2$, proving (C10).

On the continuous sphere $y^{\mathsf{T}} y = 64$, equation (C1) has minimum

$$\min_{\|y\|^2 = 64} F(\sigma, y) = 560 - 16\lambda_{\max}(\Lambda_\sigma). \tag{C11}$$

This relaxed minimum is 32 lower for σ_1 than for σ_0. The binary minimum is bounded below by the spherical one; a binary vector need not be a top eigenvector. Corollary 15 alone therefore does not prove a difference of 32, or any difference, between exact binary completion minima. A finer enclosure of ρ and one explicit completion do.

Corollary 16 (the binary completion minima differ). Every completion of σ_0 violates at least 347 of the 1120 outside lines, every completion of σ_1 violates at least 315 of them, and some completion of σ_1 violates exactly 344. Hence

$$\min_y F(\sigma_1, y) \;\leq\; 344 \;<\; 347 \;\leq\; \min_y F(\sigma_0, y). \tag{C12}$$

Proof. By (C11) and Corollary 15 the two lower bounds are $560 - 8(\rho - 1) = 568 - 8\rho$ and $560 - 8(\rho + 3) = 536 - 8\rho$. The polynomial g is increasing from 25 on, and

$$g\big(\tfrac{221}{8}\big) = -\tfrac{25539}{512} \;<\; 0 \;<\; \tfrac{3067}{64} = g\big(\tfrac{111}{4}\big)\,,$$

so $221/8 < \rho < 111/4$, hence $346 < 568 - 8\rho < 347$ and $314 < 536 - 8\rho < 315$. The number of violated lines is an integer, which gives the two lower bounds. For the upper bound let y be -1 at the 30 outside labels listed in Appendix B and $+1$ at the other 34; counting the 1120 outside lines one by one gives 344 violated lines, equivalently $y^{\mathsf{T}} \Lambda_{\sigma_1} y = 864$. ∎

Simulated annealing with 6000 restarts found a completion of σ_0 violating 368 outside lines (also listed in Appendix B). The two exact minima therefore lie in $[347, 368]$ and $[315, 344]$; the upper ends are search results, not proved optima. `scripts/cospectral_pair.py` (30 checks) rebuilds the configuration, both carriers, both degree-64 completion polynomials, the enclosure of ρ and both witnesses in exact arithmetic, without code shared with the C3 programs of Section 11.2.

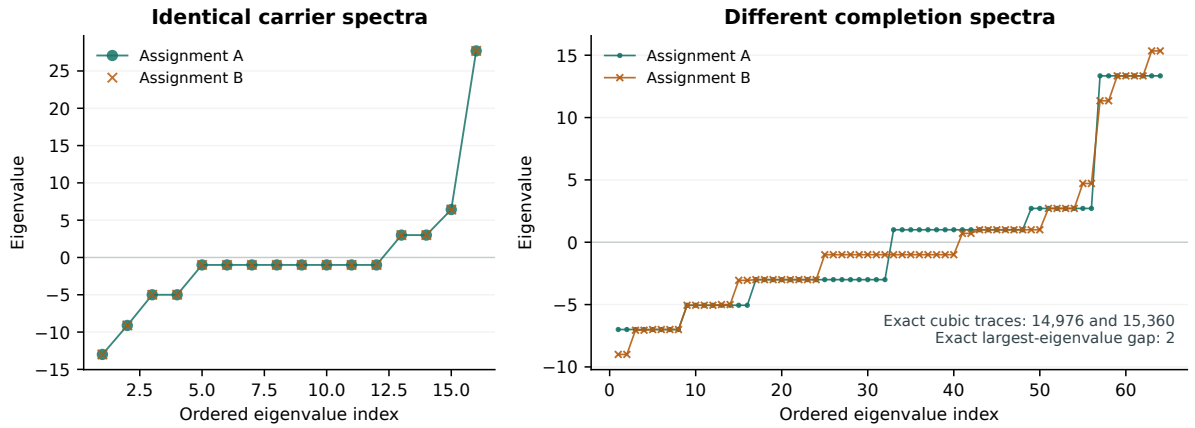


Figure 3. C3 carrier and completion spectra. Assignment A is σ_0 and B is σ_1. Numerical eigenvalue locations illustrate the certificate; exact integer identities establish carrier cospectrality, unequal completion cubic traces and the largest-eigenvalue gap. Multiplicities are retained.

The fixed-section score is also identical. The section has $1575 - 1120 = 455$ contexts. The trace identity of Section 3 gives $\beta_H = 18336/(6 \cdot 16) = 191$ for each carrier, hence $(455 - 191)/2 = 132$ violated section contexts and 323 satisfied ones. Independent integer Pauli-label phase arithmetic in `scripts/unified_section_score.py` recounts these values; the record is `results/unified-section-score.json`. Flipping label 1 of the first assignment changes its count to 147 and its score to 161, furnishing a negative control. This section-score observation is a derived and executable check, not an additional specialized Lean theorem.

8.6 Case C3 for every number of qubits from four on

Let $N \geq 3$, $m = N - 1$ and $k = 2^m$. Let $\mathcal{Q} = Q^+(2N-1, 2)$ be the set of real nonidentity N-qubit Pauli observables, $p_\star = Z \otimes I_k$, H the section (the points of $\mathcal{Q}$ that commute with $p_\star$) and $O = \mathcal{Q} \setminus H$. With R the set of real nonidentity m-qubit Pauli observables, $|R| = (k-1)(k+2)/2$,

$$H = \{p_\star\} \cup \{I \otimes P,\ Z \otimes P : P \in R\}, \qquad |H| = 1 + (k-1)(k+2), \qquad |O| = k^2. \tag{E1}$$

Carrier operators and completion matrices are defined as in Section 8.2. On m qubits write vectors of $\mathbb{F}_2^m$ as integers, $a \cdot c$ for the number of common bits, $P_{a,c} = i^{a \cdot c} X^a Z^c$, which is real exactly when $a \cdot c$ is even, and $X^a Z^c |b\rangle = (-1)^{c \cdot b} |b + a\rangle$.

Carriers from graphs. Choose a vertex $t \in \mathbb{F}_2^m$ and, for every nonzero $a \in \mathbb{F}_2^m$, a vertex u_a. The graph G on $\mathbb{F}_2^m$ has the $k - 1$ edges $\{u_a, u_a + a\}$, one for each nonzero difference. Define signs on R by

$$\varepsilon(P_{a,c}) = (-1)^{c \cdot u_a} (-1)^{(a \cdot c)/2} \quad (a \neq 0), \qquad \varepsilon(P_{0,c}) = (-1)^{c \cdot t}. \tag{E2}$$

They depend only on the edges, because $c \cdot a$ is even. Let F be a union of connected components of G not containing t. The carrier σ has $\sigma(p_\star) = 1$ and $\sigma(I \otimes P) = \sigma(Z \otimes P) = \varepsilon(P)$ for $P \in R$; the carrier τ agrees with σ except that $\tau(Z \otimes P_{a,c}) = -\varepsilon(P_{a,c})$ whenever a is the difference of an edge of F.

Theorem 17 (case C3, every $N \geq 4$). For $N \geq 4$ take $u_1 = 0$, $u_2 = u_3 = 4$, $u_a = 2$ for $a \geq 4$, $t = 2$ and $F = \{\{0, 1\}\}$. Then A_σ and A_τ have the same characteristic polynomial and

$$\operatorname{tr} \Lambda_\sigma^3 - \operatorname{tr} \Lambda_\tau^3 = -\tfrac{3}{2} k^3 = -\tfrac{3}{2} 8^{N-1}. \tag{E3}$$

So the completion spectra differ, and the strict law fails for sign carriers in case C3 for every $N \geq 4$. The two carriers differ exactly at the $k/2$ points $Z \otimes W \otimes X$ with $W \in \{I, Z\}^{\otimes(N-2)}$.

The edge $\{0, 1\}$ is a component, because the other edges $\{4, 6\}$, $\{4, 7\}$ and $\{2, 2+a\}$ with $a \geq 4$ avoid the vertices 0 and 1. On the direction $a = 1$ the condition $a \cdot c$ even means that bit zero of c vanishes, so $P_{1,c} = W \otimes X$ and $\varepsilon = +1$ there.

Proof. (i) *Blocks.* Every P_s with $s \in H$ commutes with $P_{p_\star}$, so $A_\sigma = C_\sigma \oplus D_\sigma$ on the two eigenspaces of $P_{p_\star}$, with

$$C_\sigma = \sigma(p_\star)I + \sum_{P \in R} \big(\sigma(I \otimes P) + \sigma(Z \otimes P)\big)P, \qquad D_\sigma = -\sigma(p_\star)I + \sum_{P \in R} \big(\sigma(I \otimes P) - \sigma(Z \otimes P)\big)P,$$

as in (C4) and [J3, Theorem B3]. Each $P \in R$ lies in exactly one block, with coefficient ± 2.

(ii) *Character sums.* Since $\varepsilon(P_{a,c})P_{a,c} = (-1)^{c \cdot u_a} X^a Z^c$, and the vectors c with $a \cdot c$ even form the hyperplane orthogonal to a,

$$\sum_{c \,:\, a \cdot c \text{ even}} (-1)^{c \cdot u_a} X^a Z^c = \tfrac{k}{2}\big(|u_a + a\rangle\langle u_a| + |u_a\rangle\langle u_a + a|\big) \quad (a \neq 0), \qquad \sum_{c \neq 0} (-1)^{c \cdot t} Z^c = k\,|t\rangle\langle t| - I. \tag{E4}$$

The first sum sends $|b\rangle$ to $\sum_c (-1)^{c \cdot (u_a + b)} |b + a\rangle$, and a character summed over a hyperplane gives $k/2$ if it is trivial there, that is if $b + u_a \in \{0, a\}$, and 0 otherwise.

(iii) *The blocks are graphs.* With Adj the adjacency matrix, (i) and (ii) give

$$\begin{aligned} C_\sigma &= -I + 2k\,|t\rangle\langle t| + k\,\mathrm{Adj}(G), & D_\sigma &= -I, \\ C_\tau &= -I + 2k\,|t\rangle\langle t| + k\,\mathrm{Adj}(G - F), & D_\tau &= -I + k\,\mathrm{Adj}(F). \end{aligned} \tag{E5}$$

(iv) *Equal carrier spectra.* Let V_2 be the vertex set of the components in F and V_1 its complement, which contains t. Then $2|t\rangle\langle t| + \mathrm{Adj}(G) = M_1 \oplus M_2$ with M_1 supported on V_1 and $M_2 = \mathrm{Adj}(F)$ on V_2. In units of k and after adding I, the pair of blocks is $(M_1 \oplus M_2,\, 0)$ for σ and $(M_1 \oplus 0,\, 0 \oplus M_2)$ for τ. Both have the union of spectra $\operatorname{spec} M_1 \uplus \operatorname{spec} M_2 \uplus \{0^k\}$.

(v) *The completion.* An outside observable anticommutes with $P_{p_\star}$, so it is off-diagonal with a real $k \times k$ block M_p, and the k^2 blocks M_p are a basis of the real $k \times k$ matrices. A section point s commuting with $p \in O$ gives exactly one outside context $\{s, p, q\}$ with $P_s P_p = s_\ell P_q$, every such context arises once, and $p_\star$ commutes with no outside point. As in (C6) and [J3, Theorem A], Λ_σ is therefore the matrix of $M \mapsto \frac{1}{2}(C_\sigma M + M D_\sigma)$ in the basis of the M_p. Its eigenvalues are $\frac{1}{2}(\gamma_i + \delta_j)$ for $1 \leq i, j \leq k$, where γ_i and δ_j are the eigenvalues of C_σ and D_σ, so

$$\operatorname{tr} \Lambda_\sigma^3 = \tfrac{1}{8}\big(k \operatorname{tr} C_\sigma^3 + 3 \operatorname{tr} C_\sigma^2 \operatorname{tr} D_\sigma + 3 \operatorname{tr} C_\sigma \operatorname{tr} D_\sigma^2 + k \operatorname{tr} D_\sigma^3\big). \tag{E6}$$

(vi) *The difference.* For both carriers $\operatorname{tr} C = k$ and $\operatorname{tr} D = -k$, and $\operatorname{tr} C^3 + \operatorname{tr} D^3 = \operatorname{tr} A^3$ is common by (iv). Pauli orthogonality gives $\operatorname{tr} C^2 = k + 4k\, r_C$ and $\operatorname{tr} D^2 = k + 4k\, r_D$, where r_C and r_D count the observables of R carried by each block and $r_C + r_D = |R|$. Hence, for every sign carrier with $\sigma(p_\star) = 1$,

$$\operatorname{tr} \Lambda_\sigma^3 = \tfrac{k}{8} \operatorname{tr} A_\sigma^3 - \tfrac{3}{2} k^2 |R| + 3k^2\, r_D(\sigma), \tag{E7}$$

and two cospectral such carriers have completion cubic traces that differ by $3k^2\big(r_D(\sigma) - r_D(\tau)\big)$. Here $r_D(\sigma) = 0$ and $r_D(\tau) = k/2$. ∎

Equation (E7) is the form of (C8) and (C9) for every N. The use of a cubic trace to separate completion spectra is that of [J3, Lemma D (D1)], which treats cases C1 and C6.

Corollary 18 (case C11). The strict law fails for sign carriers in case C11 for every $N \geq 4$.

Proof. The cone lift [J3, Lemma D (D2)] turns a pair refuting the strict law in case C3 on N qubits into such a pair in case C11 on $N+1$ qubits: add a qubit, let e be Z on it, keep the polar point, give e the value $+1$, and give s and $e+s$ the value $\sigma(s)$. The carrier spectrum becomes $2\,\mathrm{spec}\,A + 1$ together with D copies of -1, and the completion spectrum $2\,\mathrm{spec}\,\Lambda + 1$ together with $|O|$ copies of -1. Apply this to Theorem 17, and for four qubits to the three-qubit pair of Remark (a) below. ∎

For the lifted pair $\mathrm{tr}\,\Lambda'^3 = 8\,\mathrm{tr}\,\Lambda^3 + 12\,\mathrm{tr}\,\Lambda^2 + 6\,\mathrm{tr}\,\Lambda$, and $\mathrm{tr}\,\Lambda^2$ and $\mathrm{tr}\,\Lambda$ agree for the two base carriers, so the lifted difference is eight times the base difference: $-12k^3$, $k = 2^{N-1}$, for the carriers of Theorem 17 on $N \geq 4$ qubits, and $8 \cdot (-288) = -2304$ for the three-qubit pair of Remark (a).

Remarks. (a) For $N = 3$ a single edge cannot be a component away from t, but the triangle $\{0, 1, 3\}$ with $t = 2$ ($u_1 = 0$, $u_2 = 1$, $u_3 = 0$) is, and the same proof gives a pair with difference -288. Three-qubit C3 pairs are reported in [J3], in a data set that is not public; this is the instance produced by the construction. (b) For two cospectral sign carriers with the value $+1$ at $p_\star$, (E7) shows that the completion cubic traces differ exactly when the numbers of observables P with $\sigma(I \otimes P) \neq \sigma(Z \otimes P)$ differ; for value -1 the same holds with the opposite sign. For the pair of Section 8.2 these numbers are 0 and 2, which gives $3 \cdot 64 \cdot (0 - 2) = 14976 - 15360$. (c) The largest completion eigenvalues of the two carriers of Theorem 17 differ by exactly $k/2$. (d) Small changes of the all-plus carrier do not suffice at five qubits. Among the 10 908 676 sign carriers with $+1$ at $p_\star$ that differ from it in at most three block observables (each by a sign flip in C or a move to D with either sign), no two are cospectral with different block spectra. At four qubits, among 182 176 such carriers, exactly four cospectral classes contain different block spectra, three of them with different r_D, and the pair of Section 8.2 lies in one of the three. The search (`scripts/c3_sparse_search.py`) is exhaustive and exact; it says nothing about pairs with only one member in this set.

Table 5. Explicit instances of Theorem 17 (the triangle for $N = 3$) and of Corollary 18.

Case	Qubits	Section, outside points, outside contexts	$\mathrm{tr}\,\Lambda^3$ of the two carriers
C3	3	19, 16, 72	0 and 288
C3	4	71, 64, 1120	15744 and 16512
C3	5	271, 256, 17280	649728 and 655872
C3	6	1055, 1024, 269824	23021568 and 23070720
C11 (cone lift)	4	39, 32, 304	1728 and 4032
C11 (cone lift)	5	143, 128, 4544	152832 and 158976
C11 (cone lift)	6	543, 512, 69376	5612544 and 5661696

At four qubits the C3 carriers differ at $ZIIX$, $ZIZX$, $ZZIX$ and $ZZZX$. Every row is checked on the complete configuration built from the definitions (Section 11.3).

8.7 All nine vertex cases

We use the notation of [J3]: X is the configuration of the case, $q_c = q_0 + \langle c, \cdot \rangle$ with $q_0(a \mid c) = |a \wedge c|$ modulo two, $Q_c = \{v \neq 0 : q_c(v) = 0\}$ (hyperbolic for $q_0(c) = 0$, elliptic for $q_0(c) = 1$),

d the point that defines the hyperplane, $H = X \cap d^{\perp}$ the section and $O = X \setminus H$. For a line $L = \{s, p, q\}$, $P_s P_p P_q = s_L I$. Carrier operators and completion matrices are defined as in Section 8.2.

Theorem 19 (Conjecture E). In each of the nine vertex cases and for every $N \geq 4$ there are sign carriers σ and τ on the section whose carrier operators have the same characteristic polynomial and whose completion matrices have different cubic traces, hence different spectra. In C1, C3, C6 and C11 this also holds for $N = 3$. Conjecture E is therefore true.

Table 6. The constructions of Theorem 19.

Cases	Construction	$\operatorname{tr}\Lambda_\sigma^3 - \operatorname{tr}\Lambda_\tau^3$
C1, C3, C5	graphs over $\mathbb{C}$, $\mathbb{R}$, $\mathbb{H}$; a component moves from the block $P_d = +1$ to the block $P_d = -1$	$-3\kappa k^2 r$, $k = 2^{N-1}$
C12, C15	the complex graph on $N-2$ qubits; a component moves from the block $P_e = +1$ to the block $P_e = -1$	$6k_b^2 r$, $k_b = 2^{N-2}$
C6, C11, C14	the pair swap: the whole base moves from one coupled pair to the other	$-24\kappa k_b^2 \lvert Y_0\rvert$
C8	C1 on $N-1$ qubits, through [J3, Remark 2]	that of C1 on $N-1$ qubits

Here $\kappa = 2$ for the complex cases C1 and C6 and $\kappa = 1$ otherwise, $r > 0$ is the number of section points at which σ and τ differ, and $|Y_0|$ is the number of points of the base Y_0 of the section, a nondegenerate W, Q^+ or Q^- on $N-2$ qubits (see the pair swap below), which is positive for $N \geq 4$.

Two facts from [J3, Theorems A and B3] are used: Theorem A identifies Λ_σ with Jordan multiplication by A_σ on the span of the outside observables, and Theorem B3 decomposes that operator. Let χ run over the joint eigenspaces E_χ of the central Paulis of the case and let A_χ be the block of A_σ on E_χ; a coupled pair is two eigenspaces that differ only in the eigenvalue of P_d.

1. *Preserved pairs* (cases C1, C3, C5, C6, C11 and C14). On a coupled pair, Λ_σ acts on a real space of dimension κk_χ^2, where k_χ is the common dimension of the two eigenspaces ($k = 2^{N-1}$ in C1, C3 and C5, $k_b = 2^{N-2}$ in C6, C11 and C14), and its eigenvalues there are those of $M \mapsto \frac{1}{2}(A_\chi M + M A_{\chi'})$, that is $\frac{1}{2}(\gamma_i + \delta_j)$ for $1 \leq i, j \leq k_\chi$ with γ_i and δ_j the eigenvalues of A_χ and $A_{\chi'}$, each taken κ times. Hence

$$\operatorname{tr}\Lambda^3\big|_{\text{pair}} = \tfrac{\kappa}{8}\big(k_\chi \operatorname{tr} A_\chi^3 + 3 \operatorname{tr} A_\chi^2 \operatorname{tr} A_{\chi'} + 3 \operatorname{tr} A_\chi \operatorname{tr} A_{\chi'}^2 + k_\chi \operatorname{tr} A_{\chi'}^3\big). \tag{E8}$$

2. *Swapped pairs* (cases C12 and C15). The block on one P_e-eigenspace is $A' \oplus \bar{A}'$ for a Hermitian A' of size $k_b = 2^{N-2}$ with spectrum μ, and the completion restricted to it has the eigenvalues $\frac{1}{2}(\mu_i + \mu_j)$ for $i \leq j$ in case C12 or $i < j$ in case C15, each twice. With $p_j = \operatorname{tr} A'^j$,

$$\operatorname{tr}\Lambda^3\big|_{\text{block}} = \tfrac{1}{4}\big((k_b \pm 4)\, p_3 + 3\, p_1 p_2\big), \qquad + \text{ for C12, } - \text{ for C15.} \tag{E9}$$

In cases C1, C3 and C5 every sign carrier places each block observable in exactly one of the two blocks, with coefficient ± 2; the constant term is ± 1, and Pauli orthogonality gives $\operatorname{tr} A_\chi^2 = k(1 + 4r_\chi)$ with r_χ the number of block observables carried by A_χ.

C1, C3, C5: graphs over $\mathbb{C}$, $\mathbb{R}$ and $\mathbb{H}$. Put $d = Z$ on qubit $N-1$ and let the other $N-1$ qubits form the block, so that $H = \{d\} \cup \{I \otimes P,\, Z \otimes P : P \in R\}$ with R the block observables in

X and $A_\sigma = C \oplus D$ as in Section 8.6. Case C3 ($X = Q^+$) is Theorem 17. In case C1 ($X = W$), R contains all observables. For $a \neq 0$ and $a \cdot c$ odd put $\varepsilon(P_{a,c}) = (-1)^{c\cdot u_a}(-1)^{(a\cdot c-1)/2}$, so that $\varepsilon(P_{a,c})P_{a,c} = (-1)^{c\cdot u_a}\, iX^aZ^c$, and keep (E2) where $a \cdot c$ is even. Summing over the affine hyperplane $\{c : a \cdot c \text{ odd}\}$,

$$\sum_{c\,:\,a\cdot c \text{ odd}} (-1)^{c\cdot u} X^a Z^c = \tfrac{k}{2}\big(|u+a\rangle\langle u| - |u\rangle\langle u+a|\big), \tag{E10}$$

which for $u = u_a$ is $\frac{k}{2}J_a$ with the antisymmetric edge $J_a = |u_a + a\rangle\langle u_a| - |u_a\rangle\langle u_a + a|$. With the real edge $E_a = |u_a + a\rangle\langle u_a| + |u_a\rangle\langle u_a + a|$ of (E4) and every observable in C, therefore, $C = -I + 2k|t\rangle\langle t| + k\sum_{a\neq 0}(E_a + iJ_a)$ and $D = -I$: a Hermitian weighted graph with a real edge E_a and an imaginary edge iJ_a on each pair $\{u_a, u_a + a\}$. In case C5 ($X = Q_c$ with $c = Y$ on qubit 0, which is elliptic with $q_c(d) = 0$) qubit 0 of the block carries a quaternionic structure and the other $N-2$ qubits carry the graph. Then R consists of the $P'' \otimes I$ with P'' real and the $P'' \otimes X$, $P'' \otimes Y$, $P'' \otimes Z$ with P'' imaginary (qubit 0 is the rightmost factor). With the signs of case C1, $\varepsilon(P'' \otimes I) = \varepsilon(P'')$ and each of $P'' \otimes X$, $P'' \otimes Y$, $P'' \otimes Z$ also gets the sign $\varepsilon(P'')$; the block is a quaternion-Hermitian graph with edges in the four quaternion units for each direction. In each case the observables of one direction and one edge type form a unit. Moving the units of a union of components away from t to D (that is, $\tau(Z \otimes P) = -\sigma(Z \otimes P)$ on them) leaves the carrier spectrum unchanged as in step (iv), and (E8) with $\operatorname{tr} C = k$, $\operatorname{tr} D = -k$ gives the difference $-3\kappa k^2 r_D(\tau) < 0$. With at least three graph qubits take the graph of Theorem 17 for every edge type and move the units of direction 1; with two graph qubits take the triangle of Remark (a) and move all units. This gives C1 and C3 for $N \geq 3$ and C5 for $N \geq 4$, since C5 has one graph qubit fewer. With a single graph qubit every edge contains t, in agreement with the strict law holding in C5 at $N = 3$ [J3].

C12 and C15: the complex graph between the two P_e-blocks. Put $e = Z$ on qubit $N-1$, $d = Y$ on qubit $N-2$, and let the other $N-2$ qubits form the base. For C12, $X = Q^+ \cap e^\perp$; for C15, $X = Q_c \cap e^\perp$ with $c = Y$ on base qubit 0. The section is

$$H = \{e\} \cup \{Z_e^{\alpha} Y_d^{q_c(w)} \otimes w : \alpha \in \{0,1\},\ w \neq 0 \text{ a base observable}\}.$$

On the eigenspace $Z_e = s$, $Y_d = +1$ the carrier acts as the Jordan block $A'_s = s\,\sigma(e) + \sum_w \big(\sigma(I \otimes w) + s\,\sigma(Z_e \otimes w)\big)P_w$, and on $Y_d = -1$ as its conjugate by the antiunitary of [J3, equation (1)], with the same spectrum. These are the complex blocks of C1 on $N-1$ qubits, with the graph on the $N-2$ base qubits. Take the complex graph signs of C1 on the base, $\sigma(e) = 1$, and every observable in the block $s = +1$; the graph is the triangle of Remark (a) when the base has two qubits and the graph of Theorem 17 when it has more. With $k_b = 2^{N-2}$ and Γ the Hermitian graph matrix $\sum_a (E_a + iJ_a)$ of the edges of G, $A'_+ = -I + 2k_b|t\rangle\langle t| + k_b\Gamma_G$ and $A'_- = -I$, because e contributes I and the observables with $a = 0$ contribute $2k_b|t\rangle\langle t| - 2I$. Moving the units of a component F away from t (all units for the triangle, those of direction 1 otherwise) to the block $s = -1$ gives $A'_+ = -I + 2k_b|t\rangle\langle t| + k_b\Gamma_{G-F}$ and $A'_- = -I + k_b\Gamma_F$, and the union of the two Jordan spectra is unchanged. In (E9), $p_1(A'_\pm) = \pm k_b$ for both carriers, and $p_2(A'_+) + p_2(A'_-)$ and $p_3(A'_+) + p_3(A'_-)$ are fixed by the carrier spectrum. Summed over the two blocks, (E9) is therefore a constant minus $\frac{3}{2}k_b\,p_2(A'_-)$, and $p_2(A'_-)$ increases by $4k_b r$ from σ to τ, so the difference is $6k_b^2 r > 0$ in both cases. The graph needs two base qubits, hence $N \geq 4$; with one base qubit every edge contains t, in agreement with the strict law holding in C12 and C15 at $N = 3$.

C6, C11, C14: the pair swap. Let X be the cone over $Y \in \{W, Q^+, Q^-\}$ on $N-1$ qubits with vertex $e = Z$ on qubit $N-1$, and $d = Z$ on qubit $N-2$ (C6, C11, C14 respectively; for C14, $c = Y$ on qubit 0). Then

$$H = \{e, d, e+d\} \cup \{Z_e^{\alpha} Z_d^{\beta} \otimes w : \alpha, \beta \in \{0,1\},\ w \in Y_0\},$$

where the base Y_0 is a nondegenerate W, Q^+ or Q^- on the remaining $N-2$ qubits. Put $\sigma = \tau = 1$ on e, d and $e+d$, and at $Z_e^{\alpha} Z_d^{\beta} \otimes w$

$$\sigma = (-1)^{\alpha+\beta}, \qquad \tau = (-1)^{\beta}. \tag{E11}$$

The blocks on $(Z_e, Z_d) = (\pm, \pm)$ have constant terms $3, -1, -1, -1$ for $(+,+)$, $(+,-)$, $(-,+)$, $(-,-)$. Under σ every base observable sits in the block $(-,-)$ with coefficient 4, under τ in the block $(+,-)$, and both blocks have the constant -1. So $A_\tau = U A_\sigma U^\dagger$ for the controlled-NOT U that flips Z_e when $Z_d = -1$, and the carrier spectra agree. The coupled pairs are $\{(+,+),(+,-)\}$ and $\{(-,+),(-,-)\}$, so the base moves from one pair to the other. With $T_0 = \sum_{w \in Y_0} P_w$ and $k_b = 2^{N-2}$, (E8) gives

$$\operatorname{tr} \Lambda_\sigma^3 - \operatorname{tr} \Lambda_\tau^3 = \kappa k_b \sum_{g \in \operatorname{spec} T_0} \big((2g-1)^3 - (2g+1)^3\big) + 2\kappa k_b^2 = -24 \kappa k_b \operatorname{tr} T_0^2 = -24 \kappa k_b^2 |Y_0|. \tag{E12}$$

This is negative as soon as the base is not empty: for every $N \geq 3$ in C6 and C11, and for every $N \geq 4$ in C14. At $N = 3$ the base of C14 is the empty elliptic quadric $Q^-(1,2)$, in agreement with the strict law holding there. The construction does not use Theorem 17; the cone lifts of the pairs of C1, C3 and C5 on $N-1$ qubits give a second family, which is also checked.

C8: through C1. Let $X = Q^+ \cap e^\perp$ with the nucleus $e = Y$ on qubit $N-1$, and $d = Z$ on qubit $N-2$. Every point of X is $I \otimes P'$ with P' real or $Y \otimes P'$ with P' imaginary, for a unique nonidentity Pauli P' on the other $N-1$ qubits. Two points commute exactly when their P' do, and on every line the factors Y occur an even number of times, so the line signs are those of $W(2N-3, 2)$. On the eigenspace $Y = +1$ the carrier operator is the C1 carrier operator on $N-1$ qubits for the transported carrier, and on $Y = -1$ its complex conjugate. So $\operatorname{spec} A_\sigma$ is twice the C1 spectrum on $N-1$ qubits and Λ_σ is the C1 completion matrix on $N-1$ qubits; this is [J3, Remark 2] made explicit. The C1 pairs on $N-1 \geq 3$ qubits give C8 for every $N \geq 4$. ■

Remarks. (e) The constructions need $N \geq 4$ exactly in the five cases C5, C8, C12, C14 and C15, in which [J3] finds the strict law true for all sign carriers at $N = 3$ by exhaustive computation. (f) For C1 and C6 the theorem gives an explicit proof, without computer search, of [J3, Corollary 3], whose base pairs are in a data set that is not public. (g) In the cone cases the two carrier operators are conjugate by a controlled-NOT. That gate fixes d and maps the section onto itself, but it maps $e^\perp$ to $(e+d)^\perp$; an outside point p has $\langle p, e+d \rangle = 1$, so its image lies outside $e^\perp$ and hence outside X. The gate therefore does not carry one completion problem to the other.

Table 7. Exact values for the pairs of Table 6, three to six qubits (`scripts/conjecture_e.py`).

Case	Qubits	Section, outside points, outside lines	$\operatorname{tr} \Lambda_\sigma^3$, $\operatorname{tr} \Lambda_\tau^3$	Difference
C1	3	31, 32, 240	−192, 960	−1152
C1	4	127, 128, 4032	57600, 60672	−3072
C1	5	511, 512, 65280	2491392, 2515968	−24576
C1	6	2047, 2048, 1047552	90181632, 90378240	−196608
C3	3	19, 16, 72	0, 288	−288
C3	4	71, 64, 1120	15744, 16512	−768

Case	Qubits	Section, outside points, outside lines	$\operatorname{tr}\Lambda_\sigma^3$, $\operatorname{tr}\Lambda_\tau^3$	Difference
C3	5	271, 256, 17280	649728, 655872	−6144
C3	6	1055, 1024, 269824	23021568, 23070720	−49152
C5	4	55, 64, 864	−1152, 3456	−4608
C5	5	239, 256, 15232	219648, 231936	−12288
C5	6	991, 1024, 253440	9750528, 9848832	−98304
C6	3	15, 16, 56	−288, 288	−576
C6	4	63, 64, 992	8064, 19584	−11520
C6	5	255, 256, 16256	732672, 926208	−193536
C6	6	1023, 1024, 261632	36255744, 39389184	−3133440
C8	4	31, 32, 240	−192, 960	−1152
C8	5	127, 128, 4032	57600, 60672	−3072
C8	6	511, 512, 65280	2491392, 2515968	−24576
C11	3	11, 8, 20	−96, 96	−192
C11	4	39, 32, 304	1344, 4800	−3456
C11	5	143, 128, 4544	129792, 183552	−53760
C11	6	543, 512, 69376	5864448, 6693888	−829440
C12	4	31, 40, 380	1776, 624	1152
C12	5	127, 144, 5112	116640, 113568	3072
C12	6	511, 544, 73712	3554880, 3530304	24576
C14	4	23, 32, 176	−960, 960	−1920
C14	5	111, 128, 3520	37632, 79104	−41472
C14	6	479, 512, 61184	3357696, 4088832	−731136
C15	4	31, 24, 132	720, −432	1152
C15	5	127, 112, 3080	46944, 43872	3072
C15	6	511, 480, 57360	2211264, 2186688	24576

Every difference equals the formula of its row in Table 6. The C11 rows use the pair swap and therefore differ from the cone lifts of Table 5.

8.8 Why the spectral floor survives

The floor uses spectral moments of an assignment on a whole chosen configuration to certify a universal inequality. The strict law asks whether a section carrier's eigenvalues encode a different operator. The block partition explains why the first method remains valid when the second inference fails: in every construction above the carriers have the same spectrum, and only the allocation of that spectrum between eigenspaces of a central Pauli changes. Neither the known degree 315 of $Q^+(7,2)$ nor the degree 75 of S_B is altered by these completion examples.

9. Formal verification and its boundary

The sources are in `lean/HoggarBound/` (Lean 4.35.0, Mathlib v4.35.0).

9.1 Floor and saturation statements

module	content
`Moment.lean`	`sum_cube_le` (the scalar inequality); `trace_pow_eq_sum_eigenvalues_pow` ($\operatorname{tr} A^k = \sum \lambda_i^k$ for Hermitian A); `trace_cube_le`
`Pauli.lean`	bit vectors `Fin N → ZMod 2`; Weyl operators and `pauli u` as explicit complex matrices; `pauli_mul`, `context_product` ($P_u P_v P_{u+v} = \tau(u,v) I$), `pauli_comm`; traces of one, two and three Pauli operators; $\tau = \pm 1$ on commuting pairs
`Bound.lean`	`flags` (ordered commuting pairs with product in S: six per context), `beta6`, the operator `opA`; `trace_opA_cube` ($\operatorname{tr} A^3 = D\,\beta_6$); `hoggar_bound`
`Counting.lean`	`violated`, `beta6_eq_card`; the flag count of the full space for every N (`card_flags_all`); `violatedLines` and `card_violated_le`; `four_qubit_contexts`
`Corollaries.lean`	`full_space_bound`, `full_space_fraction` (all N); `three_qubit_contexts`, `five_qubit_contexts`
`Quadric.lean`	the real quadric; its point and flag counts for every N by character sums; `four_qubit_hyperbolic_contexts`, `five_qubit_hyperbolic_contexts`
`MaskConfig.lean`	configurations and assignments given by bit masks; `card_flags_cfg`, `card_violated_cfg` (their flag counts as sums the kernel can evaluate); `degree_of_certificate`
`FlatData.lean`, `SaturatedFive.lean`	generated: the masks of the first six rows of Table 1, and the five-qubit counts in 32 blocks
`Saturated.lean`	`pivot_degree`, `cieslinski5_degree`, `quadric_degree`, `hoggar_degree`, `ccz_degree`, `lly_degree`
`MaskRows.lean`	packed sets of codes; `flagRow_eq_fast`, `violRow_eq_fast` (a row of either count, evaluated with a bounded number of operations on large numbers, for every N)
`PhaseData.lean`, `SaturatedPhaseFive.lean`, `SaturatedPhaseSeven.lean`	generated: the masks of Φ_5 and Φ_7; `phase5_degree`, `phase7_degree`
`CarrierFamily.lean`	case C3 for every number of qubits from four on (Section 9.3): `c3_strict_law_fails_general`, `c3_all_qubits_completion`

The main theorem, as checked:

```
theorem hoggar_bound (hN : 1 ≤ N) (S : Finset (Pt N)) (h0 : (0 : Pt N) ∉ S) (f : Pt N → ℝ)
    (hf : ∀ u ∈ S, f u = 1 ∨ f u = -1) :
    ((2 : ℝ) ^ N - 1) * beta6 S f
      ≤ ((2 : ℝ) ^ N - 2) * S.card * Real.sqrt (((2 : ℝ) ^ N - 1) * S.card)
```

Here `beta6 S f` is six times $\beta_S(f)$, a sum over ordered pairs, and the sign of a context is by definition the real part of the scalar $\tau(u,v)$ with `pauli u * pauli v * pauli (u + v) = tau u v • 1`. Table 8 lists the corollaries.

Table 8. Formal floor and exact-degree statements.

Lean theorem	statement
`full_space_fraction`	for all $N \geq 1$: #violated flags $\geq \left(\frac{1}{2} - \sqrt{2^N+1}/(2^N+2)\right)$ #flags

Lean theorem	statement
`three_qubit_contexts`	every assignment of the 63 observables violates at least 63 contexts
`four_qubit_contexts`	every assignment of the 255 observables violates at least 1451 contexts
`five_qubit_contexts`	every assignment of the 1023 observables violates at least 28786 contexts
`four_qubit_hyperbolic_contexts`	every assignment of the 135 real observables violates at least 315 contexts
`five_qubit_hyperbolic_contexts`	every assignment of the 527 real observables violates at least 6426 contexts
`pivot_degree`	for the 135 observables of S_B: every assignment violates at least 75 contexts, and some assignment violates exactly 75
`cieslinski5_degree`	for the 496 observables of S_5: at least 480, and exactly 480 for some assignment
`phase5_degree`	the same statement for the 496 observables in the support of the circuit state Φ_5
`phase7_degree`	for the 8128 observables in the support of the circuit state Φ_7: at least 729344, and exactly 729344 for some assignment
`quadric_degree`, `hoggar_degree`	the same two-sided statements with 315 (four-qubit hyperbolic quadric) and 63 (three qubits)
`ccz_degree`, `lly_degree`	some assignment of the 28 (respectively 12) observables violates no context

`#print axioms` reports `[propext, Classical.choice, Quot.sound]` for each of them: no `sorry`, no `native_decide`, no additional axiom.

How the exact degrees are checked. The statement of `pivot_degree` is about a set of 135 explicit complex 16×16 matrices, given by a 256-bit mask: bit $a + 16c$ says whether $P_{(a|c)}$ belongs to it. `MaskConfig.lean` proves, for every N and every mask, that the number of flags of the configuration and the number of flags violated by an assignment that is itself given by a mask are double sums over codes of Boolean expressions built from bit tests, exclusive or and addition of natural numbers. The kernel evaluates these sums (`decide +kernel`); for five qubits each sum has 2^{20} terms and is evaluated in 16 blocks. `degree_of_certificate` then combines three evaluated numbers (observables, flags, flags violated by the witness) with `hoggar_bound`. That the masks are the supports and sign patterns of the states of Table 1 is checked by the programs of Section 11 in exact arithmetic; it is not needed for, and not part of, the Lean statements. For seven qubits the two sums have 2^{28} terms each. `MaskRows.lean` evaluates a row of either sum on packed sets of codes, with at most $2N$ exchanges of blocks of digits for each of the one or two translates that a row needs, bitwise operations and one remainder, and proves for every N that the result is the row; the kernel then evaluates the 2^{14} rows in 16 blocks of 1024. The degree 729 344 of S_7 is the theorem `phase7_degree`, and `phase7_contexts` gives its 2 824 192 contexts.

Floor and saturation results outside the formalization. Theorem 2 (the quaternionic form) and all elliptic-quadric numbers; Theorems 3 and 4 as equivalences, and the state, spectral-equality and geometric interpretations in Sections 4–6 (the Lean results for that part are the inequality and the two-sided degree statements); Propositions 4, 6 and 8 to 11; the enumerations of Theorems 12 and 13, and item 1 and the local numbers of item 2 of Theorem 13; the limit statement as a theorem about limits (the inequality for each N is formal); the exact values of [J1]. For the classical families beyond three qubits and $Q^+(7, 2)$, upper bounds are witnesses checked by the programs of Section 11.

Independent audit. A separate set of agent sessions, working concurrently in the same repository and without editing our files, re-checked the first six modules from source, con-

firmed the axiom report, and added two modules of its own (`IndependentContexts.lean`, `IndependentWitness.lean`). They prove that a violated context carries exactly six violated flags (used in `MaskConfig.lean`) and, by kernel evaluation of an integer certificate, that the explicit assignment "$f(a, c) = +1$ if $a = 0$ or $c = 0$ or the number of Y factors is at least 2, and -1 otherwise" violates exactly 63 contexts. Their reports are in `results/lean-audit.md`, `results/literature-audit.md` and `results/stack-independent-audit.md`, and their companion note is `paper/independent-research-note.md`. The literature audit is what first pointed us to [J1]. The modules from `MaskConfig.lean` on, and everything in Section 5, were written after that audit and are not covered by it.

9.2 The checked four-qubit C3 statement

The C3 development uses the same pinned Lean 4.35.0 and Mathlib 4.35.0 setting. It identifies the complete 71-point section and 64-point complement, proves the assignments are sign-valued, and connects the integer certificate matrices to the original complex Pauli sums and real `lineSign` completion operators. Thus its conclusion applies to the configuration of Section 8.

The main declarations in `HoggarBound.C3` are `four_qubit_C3_counterexample`, `pauli_carrier_eigenvalues_eq`, `pauli_completion_eigenvalue_multisets_ne` and `four_qubit_C3_spectral_counterexample`. They prove equal carrier characteristic polynomials and canonical Hermitian eigenvalue lists, different completion characteristic polynomials and real root multisets with multiplicities, and the exact completion cube traces 14976 and 15360.

The certificate kernel-checks $A_0M = MA_1$ and $MR = 46656I$ for explicit integer matrices. This certifies an invertible similarity over the rationals. For each completion, an intermediate square is checked entry by entry and substituted into a checked cubic trace. No external eigensolver supplies a proof premise. All 42 declarations in `lean/C3Verification.lean` passed the axiom audit with only `propext`, `Classical.choice` and `Quot.sound`; the finite checks use kernel reduction, not `native_decide`. Commands, source hashes and results are in `results/frontier-c3-lean-audit.md` and its evidence files.

9.3 Case C3 for every number of qubits from four on

`lean/HoggarBound/CarrierFamily.lean` is a separate file for the same pinned Lean and Mathlib. It imports the library modules `Pauli.lean`, `Bound.lean` and `Quadric.lean` and changes no existing file. Its main theorem, for $N = m + 1$ qubits, is

```
theorem c3_strict_law_fails_general (hm : 3 ≤ m) :
    ∃ σ τ : Pt (m + 1) → ℝ,
      (∀ s, σ s = 1 ∨ σ s = -1) ∧ (∀ s, τ s = 1 ∨ τ s = -1) ∧
      (opA (sectionPts m) σ).charpoly = (opA (sectionPts m) τ).charpoly ∧
      (completionMatrix m σ).charpoly ≠ (completionMatrix m τ).charpoly
```

with the definitions

```
def polar (m : ℕ) : Pt (m + 1) := (0, Pi.single 0 1)                -- Z on qubit 0
def sectionPts (m : ℕ) : Finset (Pt (m + 1)) :=
  (realPoints (m + 1)).filter fun s => symp s (polar m) = 0
def outsidePts (m : ℕ) : Finset (Pt (m + 1)) :=
  (realPoints (m + 1)).filter fun s => symp s (polar m) = 1
noncomputable def completionMatrix (m : ℕ) (σ : Pt (m + 1) → ℝ) :
    Matrix {p // p ∈ outsidePts m} {p // p ∈ outsidePts m} ℝ :=
  fun p q => if p.1 ≠ q.1 ∧ symp p.1 q.1 = 0
    then lineSign (p.1 + q.1) q.1 * σ (p.1 + q.1) else 0
```

`realPoints`, `symp`, `lineSign` and `opA` are those of the floor theorem: the real quadric, the symplectic form, the sign of the context `{u, v, u + v}` defined by the product of the three Pauli matrices, and the signed sum of Pauli matrices. Lean places the polar point on qubit 0, which is a relabelling of the qubits of Section 8.6. The file also proves that the definitions have the intended meaning: the polar point lies on the quadric and in the section (`polar_mem_sectionPts`); two distinct commuting outside points span a context whose third point lies in the section, and only then (`add_mem_sectionPts`, `add_mem_sectionPts_iff`); a commuting section point and outside point span a context with a second outside point (`add_mem_outsidePts`); the polar point commutes with no outside point (`symp_polar_outside`); $|O| = k^2$ and $|H| = 1+2|R| = k^2+k-1$ (`card_outsidePts`, `card_sectionPts`, `card_sectionPts_add_one`); the completion matrix is symmetric (`completionMatrix_transpose`). The companion statement `c3_all_qubits_completion` gives the difference (E3) in coordinates, and `completionMatrix_liftCarrier` identifies those coordinates with `completionMatrix`.

The 74 audited declarations of the file depend only on `propext`, `Classical.choice` and `Quot.sound`; the file has no `sorry`, `native_decide`, `axiom` declaration or `set_option` (`results/lean-c3-all-qubits-axioms.log`). Left to the reader are the identification of these definitions with the objects of [J3], the fact that equal characteristic polynomials of Hermitian or real symmetric matrices mean equal spectra with multiplicities, and the trust base: the Lean kernel, Mathlib at the pinned revision and the three standard axioms. Corollary 18, Remarks (a) to (g), Corollary 16 and Theorem 19 outside case C3 are not formalized.

Table 9. Boundaries that matter for interpreting the combined result.

Claim	Evidence boundary
Arbitrary-support spectral floor and stated finite corollaries	Lean statements about the defined Pauli/context model
Degrees 75, 480 and 729344 for the explicit masks	Lean lower bounds, witnesses and kernel-evaluated counts; state-to-mask interpretation checked separately by programs
Saturation equivalences and phase-state classification	Mathematical arguments and executable computations; not general Lean equality/classification theorems
Four-qubit C3 carrier cospectrality, unequal completion spectra and cube traces	Complete 42-declaration Lean audit, including original Pauli and context-sign bridges
Printed C3 block/carrier factorizations, largest-eigenvalue gap 2, spherical gap 32 and the binary minima of Corollary 16	Exact algebraic and executable checks; not included in those 42 Lean declarations
Case C3 for every $N \geq 4$ (Theorem 17)	Lean theorem `c3_strict_law_fails_general` about the library's Pauli matrices, real quadric and context signs; 74 audited declarations
Case C11 (Corollary 18) and the other cases of Theorem 19	Paper proofs that use [J3, Theorems A and B3, Lemma D and Remark 2]; exact computation on the complete configurations for three to six qubits; not in Lean
Physical hardware, compiler correctness, historical priority or external peer review	Not established by these theorem checks

The floor, saturation and C3 audits have different scopes. The independent audit of the original six floor modules does not retroactively cover the later saturation development, and the C3 audit does not certify the phase-state enumeration. Version 1.3.0 adds two Lean developments, each audited separately: the seven-qubit evaluation (`MaskRows.lean`, `SaturatedPhaseSeven.lean`) and case C3 for every number of qubits (`CarrierFamily.lean`). They are new files; every

Lean file of the earlier audits is byte-identical to its audited version, and no earlier theorem's hypotheses or conclusion change.

Successful CI provides a second recorded execution at commit `87f08a0`. The full verification run passed the executable checks and audits of 42 C3, 22 baseline and six auxiliary stability declarations. The separate saturation run passed its exact-computation jobs and 14-declaration audit. The archived records `results/full-ci-87-audit.json` and `results/saturation-ci-audit.json` identify the checkout, artifact digests and source hashes. These are checks of those source snapshots, not evidence of a new proof produced by combining the texts; the auxiliary stability results are not used here.

The two additions have their own recorded runs on clean Ubuntu machines. The seven-qubit run of the saturation workflow, at commit `b7b0273`, built `SaturatedPhaseSeven` in 1 minute 26 seconds with a maximum resident set of 6 271 660 KiB (6.4 GB) and passed its 13-declaration axiom audit (`results/saturation-seven-ci-audit.json`). It repeats, after a review that made one program check able to fail and corrected a comment, the first run (42791713861, commit `77a529f`), whose Lean results were the same. The workflow *Case C3 for every number of qubits* checked `CarrierFamily.lean` and enforced the allowlist on its 74 declarations at commits `5dcea35` (run) and `f294d9d` (run). The first run also repeated the C3 checks of Section 11.3 (`scripts/cospectral_family.py --six`, the tensor check, the cone lifts and the four-qubit search); the second added `scripts/conjecture_e.py` for three to five qubits (81 checks). The six-qubit step (`scripts/conjecture_e.py 6`, 36 checks) was added to the workflow in this version. At commit `6189b62`, whose programs and Lean files are those of this release, the workflow ran every step, the six-qubit one included, and passed (run; logs in `results/ci-run-42817286684/`).

12. Limitations and open problems

- Novelty is claimed to the extent of a bounded search (arXiv and Zenodo queries on 9 November 2026, the 54 queries of 10 November 2026 described in Section 1.2, and the sources listed there). The field moved quickly in October 2026, and several of the relevant preprints are less than a month old. Statements of the form "to our knowledge" mean exactly that.
- The configurations S_B, S_5 and S_7 are defined by states, not by a classical geometric construction. We do not know an infinite family of non-closed configurations that attain the floor: Proposition 9 shows that the simplest family of states stops at four qubits. In the nonzero-form odd-qubit phase-state family of constant rank $N-1$, the examples here have three, five and seven qubits; the rank-eight nine-qubit case is open. Theorem 12 is complete for its class of states; whether Table 1 is complete for all states on $N \le 5$ qubits is supported by numerical search only. Whether the states of the odd class of Proposition 6 form a single Clifford orbit was not determined.
- For seven qubits the uniqueness statement is proved only inside the family $\mathcal{F}$ of Theorem 13; outside it there is a random census. The enumeration behind it (and the one of Theorem 12) is a computation, repeated here by one program only, apart from the agreement of the count with the orbit size and the second implementation for $N \le 4$.
- Within the classical families the floor is not attained beyond the three listed cases [J2]. Exact values for five or more qubits are open except $Q^+(9,2)$ [J1]; the floor alone leaves the intervals of Table 2.
- The external exact values $d(W(7,2)) = 1575$, $d(Q^-(7,2)) = 315$ and $d(Q^+(9,2)) = 6975$ rest on computer-assisted work not formally verified or peer reviewed in [J1]. This does not include $d(Q^+(7,2)) = 315$, which has the Lean proofs described above. A formal proof of $d(W(7,2)) = 1575$ would be a natural next target; our Lean development gives the lower bound 1451 only.

- The quaternionic form (Theorem 2) and the equivalences of Theorems 3 and 4 are proved on paper and tested numerically, not formalized.
- Conjecture E is settled for sign carriers in all nine vertex cases, but only case C3 is formalized in Lean. The proofs of the other eight cases use Theorems A and B3 of [J3] (with Lemma D and Remark 2 for C11 and C8), which are results of a preprint that is not peer reviewed and that we did not formalize; their instances are checked by exact computation for at most six qubits. The carriers are particular constructions: carriers in general are not classified, and for the families the completion optima are not studied.
- For the four-qubit pair the exact binary completion minima are only enclosed, in $[347, 368]$ and $[315, 344]$; the upper ends come from a heuristic search. The completion results do not invalidate the spectral floor or revise the contextuality degrees.
- The literature comparison is bounded and qualitative. An exact finite certificate proves its stated mathematical conclusion, not that no earlier author knew it or that a venue will accept it.
- Nothing here is an experiment on quantum hardware, and nothing shows a computational advantage. The simulation demonstrates what quantum mechanics predicts and what no noncontextual model can do.

13. Research provenance, disclosure and availability

How the work was produced. Alexander S. Donahue initiated and directed the project and specified the Tsotchke stack. The floor, saturation, phase-state and associated simulation work was developed by Claude Fable 5 (Anthropic) through Claude Code. Within the author's brief that system selected problems, searched sources, derived results, wrote and ran programs and Lean proofs, and drafted the spectral-floor manuscript. Its development and verification records remain with that component.

OpenAI Codex research and coding agents performed the independent prior-art and original-floor audits, selected the finite C3 question, discovered its witnesses, developed the algebraic arguments and Lean formalization, implemented the independent checks and demonstration, and drafted the C3 manuscript and figures. Codex agents prepared the combined text of version 1.2.0, its shared notation, section map and publication layouts. That version reused the two established mathematical components without a new experimental dataset or a new unified Lean proof.

Version 1.3.0 adds results obtained on 9 and 10 November 2026 by Claude (Anthropic) agents working through Claude Code under the author's direction (the commits of Theorems 17 and 19, of the seven-qubit corrections and of this revision name Claude Opus 5.5 as co-author; those of Corollary 16 and of the first seven-qubit evaluation name only Claude): Corollary 16, the kernel evaluation of the seven-qubit counts (in a separate session), Theorem 17 with its Lean formalization, Corollary 18 and Theorem 19, with the programs of Section 11.3 and `scripts/cospectral_pair.py`. They were first recorded as addenda in `publication/` and are folded into this text by the same line of work. Before publication the C3 results were examined by separate automated passes, instances of the same AI system each started without the others' work and with one task: a re-derivation of the proof and of the tables from the text alone, a full reading of [J3] against the definitions and credits, line-by-line readings and mutation tests of the programs, the prior-art queries of Section 1.2, and audits of the Lean statements, their fidelity to the mathematics and their axioms; for the Lean part that ties the completion matrix to the library's contexts the separate audit did not complete, and that part was checked by comparing all 83 statement paragraphs with the final file and by its axiom list. Findings rated critical or major were tested again before they were accepted; among the changes they caused are Corollary 18 and the Lean part that ties the completion matrix to the library's contexts. This

is internal checking, not peer review; the record is in `publication/ADDENDUM-c3-all-qubits.md`. Theorem 19 was checked by its program and by one separate pass of the same kind, recorded in `publication/ADDENDUM-conjecture-e.md`.

AI disclosure statement. Claude and Codex were used for substantive research, software, formalization and writing, not merely language correction. The recorded checks comprise Lean kernel verification, exact and numerical programs with their stated scopes, and source comparisons by AI agents. They do not constitute independent human peer review. No completed human critical review or venue-specific final approval is asserted here; the author must accurately describe that work before making a submission that requires it. AI systems are disclosed as tools, not listed as authors.

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Data availability statement. Source, data, formal proofs, test results and the earlier manuscripts are available in Hoggar Bound Lab. Release and workflow records identify the checked snapshots; Software Heritage and Internet Archive captures are archival records of particular snapshots, not evidence of review. The addenda in `publication/` record each result of version 1.3.0 as first stated. Appendix A gives reproduction commands. No new arXiv identifier, DOI or journal decision is implied by this manuscript or its local submission packages.

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AI disclosure statement. Claude (Anthropic) and OpenAI Codex were used for substantive research, software development, formalization and manuscript preparation under the author's direction. Section 13 describes their roles and the verification performed in detail. AI systems are disclosed as tools, not authors; no independent human peer review or venue-specific final approval is claimed.

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